

4 -Maintenance operations

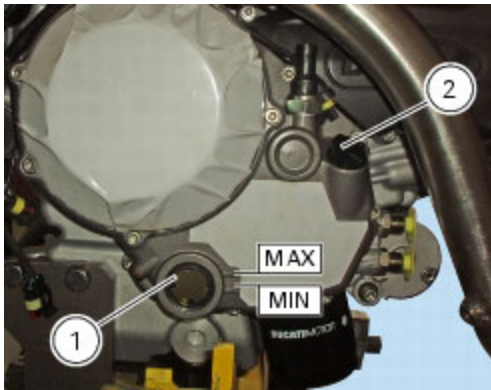
Checking the engine oil level

Check the engine oil level through the sight glass (1) on the right-hand side of the oil sump.

Stop the engine and allow a few minutes for the oil to settle to a steady level. Check the level with the motorcycle perfectly vertical and with the engine warm (but not running).

The oil must be between the MIN and MAX marks. If the level is low, top it up. Remove the filler cap (2) and top up with the recommended oil.

Refit the filler plug (2).



Changing the engine oil and filter cartridge



Note

Change the oil when the engine is warm (but with the engine stopped). In these conditions the oil is more fluid and will drain more rapidly and completely.

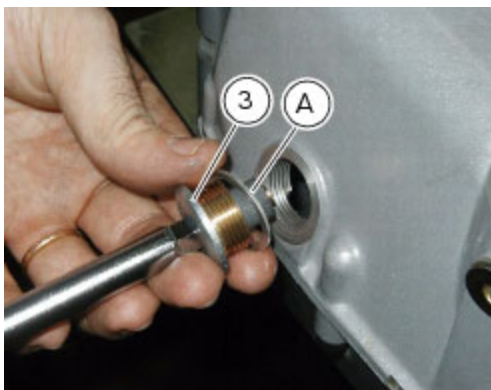
Remove the drain plug (3) with gasket (A) from the oil sump and allow the oil to drain off.



Warning

Dispose of oil and/or filter cartridges in compliance with environmental protection regulations.

Remove any metallic deposits from the end of the magnetic drain plug. Refit the drain plug complete with seal (A) to the sump.



Tighten it to the specified torque (Sect. C 3, [Engine torque settings](#)).

Remove the oil sump filter cartridge (4) using service tool **88713.2906**.



Important

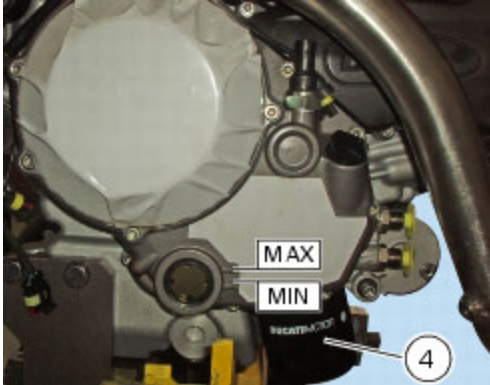
Dispose of the used cartridge, do not attempt to reutilise it.

Fit a new cartridge (4) using service tool **88713.2906**: lubricate the gasket with engine oil.



Note

It is advisable to fill the filter cartridge (4) with engine oil before installing it: this enables the recommended oil level to be maintained without topping up.



Start it in its seat and tighten it to the specified tightening torque (Sect. C 3, [Engine torque settings](#)).

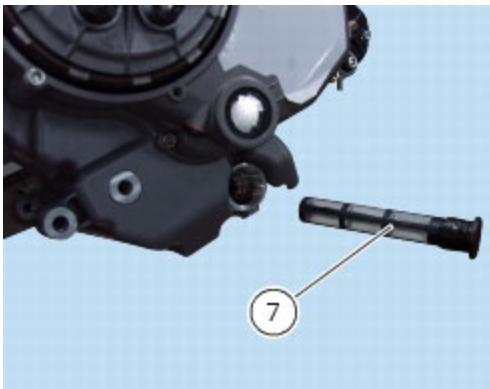
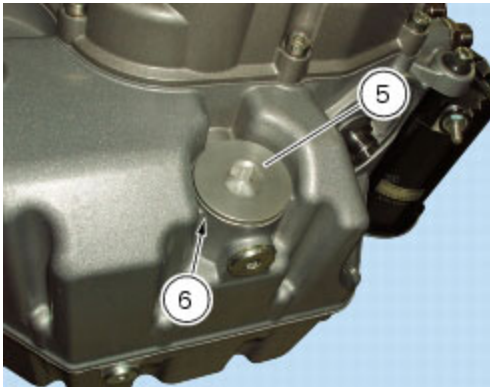
Every two oil changes, clean the oil pickup gauze filter.

Remove the outer plug (5).

Unscrew the filter (7) and withdraw it from the crankcase.

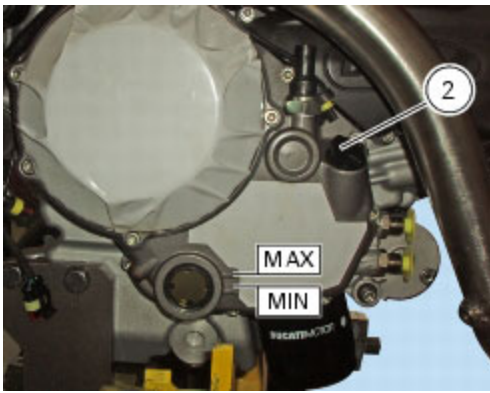
Clean the filter with petrol and compressed air. Take care not to damage the gauze.

Replace the mesh filter (7), fit the seal (6) to the plug (5) and tighten to the specified torque (Sect. C 3, [Engine torque settings](#)).



Remove the filler plug (2) and carry out refilling with the specified oil type (Sect. C 2, [Fuel, lubricants and other fluids](#)) up to reaching the notch that identifies the **MAX** level in the sight glass.

Refit the filler cap (2). Run the engine at idle speed for a few minutes.

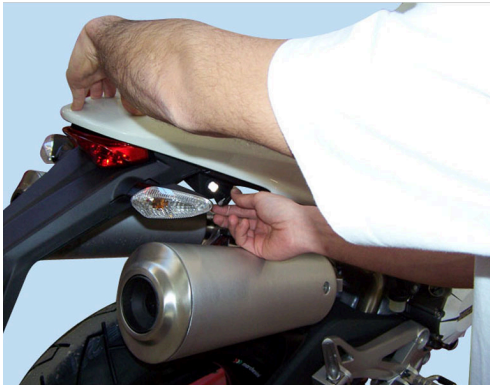


Check for oil leaks. Check that the oil pressure light on the instrument panel switches off a few seconds after the engine starts. If not, switch off the engine and trace the fault.
After a few minutes, check that the oil level is the specified one; if necessary, restore the **MAX** level.
Refit all parts removed.

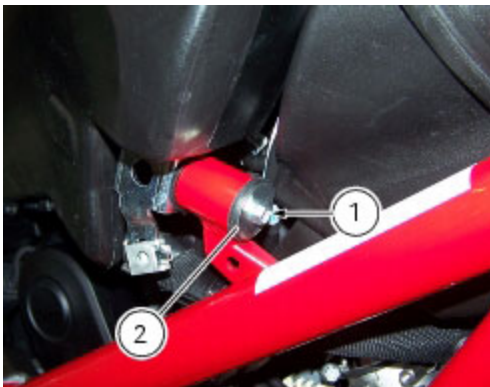
Checking valve clearances

In order to check the valve clearances, it is necessary to gain access to the four valve covers on the cylinder heads and then remove the parts listed below.

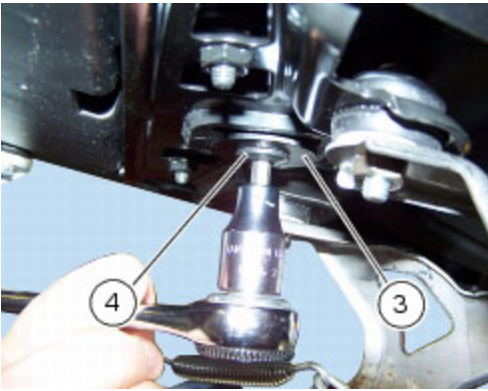
Open the lock and simultaneously apply downward pressure in the area of the catch to release the pin; Pull the seat backwards to release it from the front catches and remove it.



Remove the tank covers (Sect. E 2, [Removal of the fuel tank fairings](#)).
Unscrew the side screw (1) securing tank to the frame and recover the washer (2).



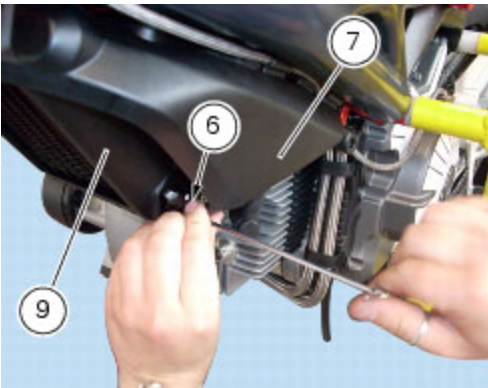
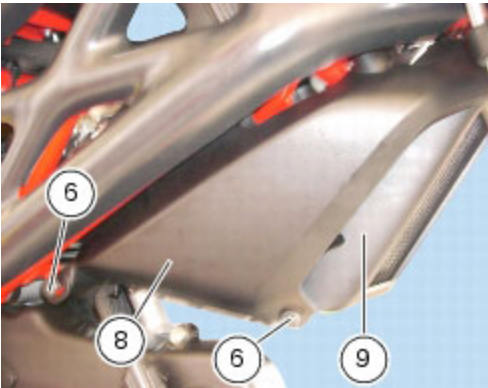
Unscrew the rear screw (3) securing tank and recover the washer (4).



Raise the tank but leave it connected to the fuel system and wiring then position it on the side, duly supported, to be able to work on the valve covers (5) of the vertical head.

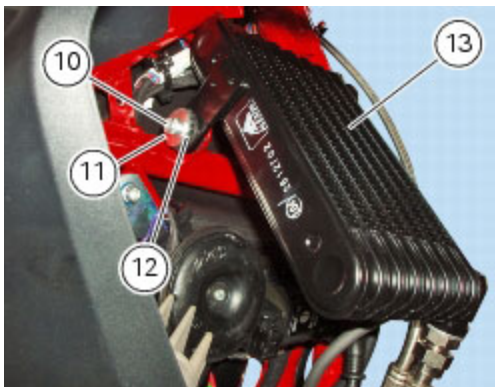


Loosen the screws (6) and remove the deflectors (7) and (8) and conveyor (9).





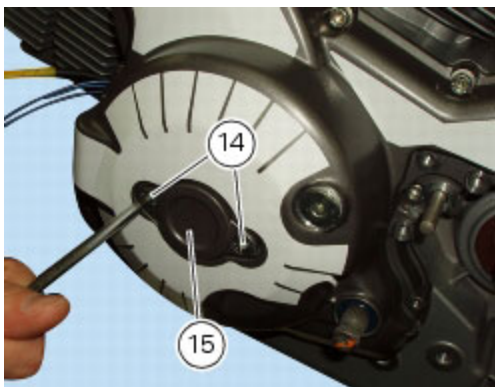
Loosen the screw (10) and recover the spacer (11) and the rubber mounting (12). Slide the oil cooler (13) off the pins of the frame, towards the clutch-side of the motorcycle.



Unscrew the bolts securing the horizontal head covers (5) and the vertical head ones then remove the covers.



Unscrew the two retaining screws (14) of the centre cap (15) over the end of the crankshaft.



Fit the tool handgrip **88713.0123** in the holes of the generator cover to be able to turn the crankshaft so that the valve on which the control is carried out is in rest position.
With the valve in the rest position, slide a feeler gauge between the opening rocker and the opening shim to measure the clearance.
The clearances must be within the specified limits:

Opening rocker arm

INTAKE: (A)

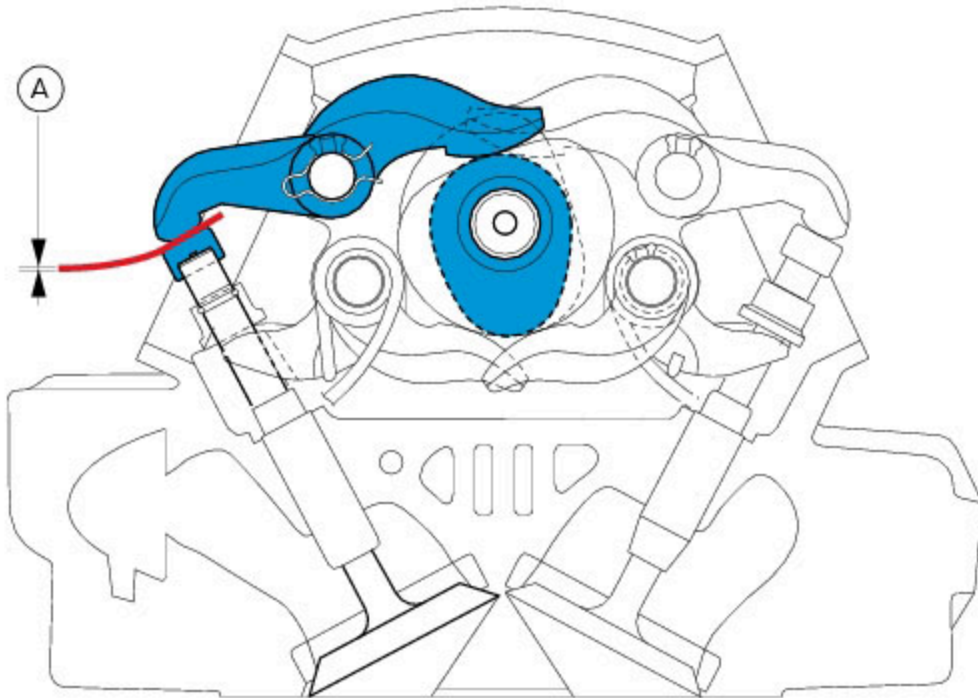
Adjusting clearance 0.10 to 0.15 mm

Checking clearance 0.05 to 0.15 mm

EXHAUST: (A)

Adjusting clearance 0.10 to 0.15 mm

Checking clearance 0.05 to 0.15 mm



With the valve in the rest position, slide a feeler gauge between the closing rocker arm and the closing shim to measure the clearance:

Closing rocker arm

INTAKE: (B)

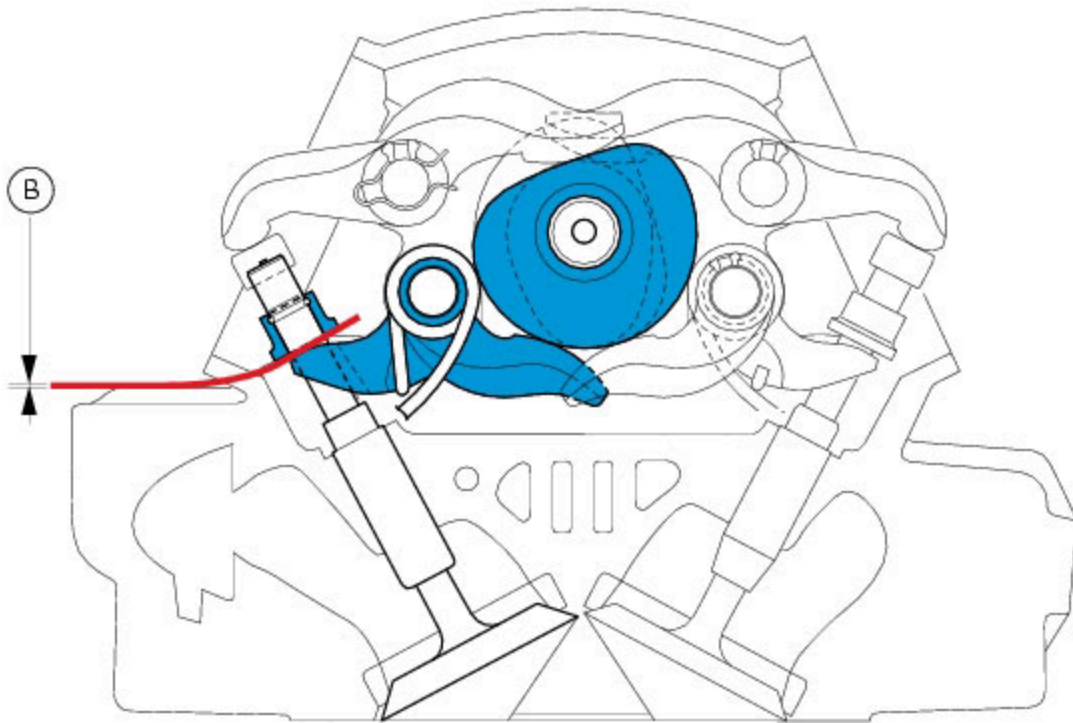
Adjusting clearance 0 to 0.05 mm

Checking clearance 0 to 0.20 mm

EXHAUST: (B)

Adjusting clearance 0 to 0.05 mm

Checking clearance 0 to 0.20 mm



If detected values exceed the specified limits, replace opening and/or closing shims, as described in paragraph "[Removal of the rocker arms, shims and valves](#)" (Sect. N 4.4), with an adequate height to obtain the specified clearance.



Note

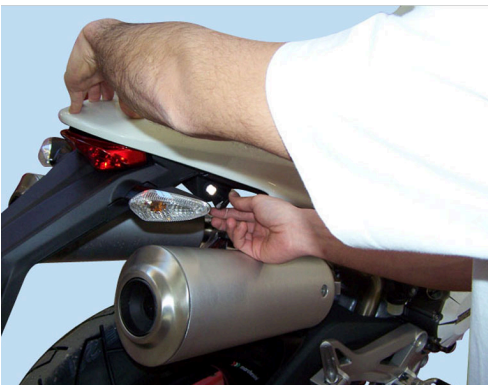
Opening rocker arm shims measuring from 1.8 to 3.45 mm and closing rocker arm shims measuring from 2.2 to 4.5 are available as spare parts. the size is punched on the shim.

Refit all the removed parts in the reverse order to that in which they were removed.

Changing and cleaning the air filters

The air filter must be replaced at the intervals described in the "[Routine maintenance table](#)" (Sect. D 3).

Open the lock and simultaneously apply downward pressure in the area of the catch to release the pin; Pull the seat backwards to release it from the front catches and remove it.

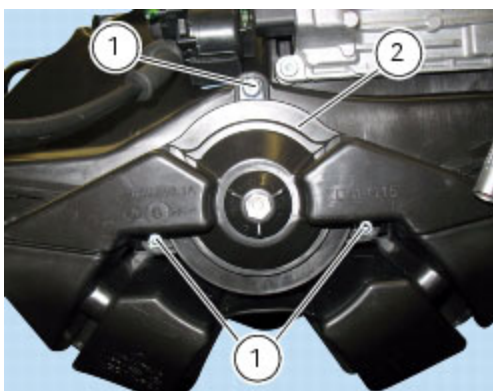


Remove the tank covers (Sect. E 2, [Removal of the fuel tank fairings](#)).

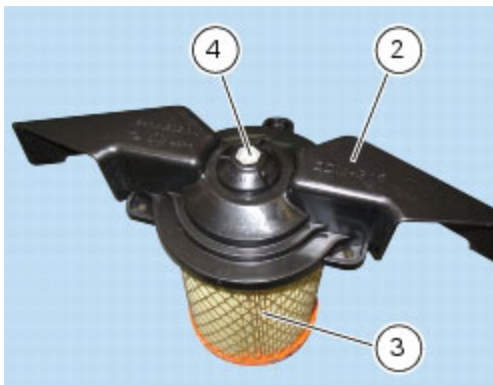
Remove the fuel tank (Sect. L 2, [Removal of the fuel tank](#)).

Unscrew the three screws (1) securing the filter cover (2).

Remove the filter cover (2) complete with the filter cartridge.



Remove the filter cartridge (3) from the cover (2) by unscrewing the screw (4).
The filter cartridge can be cleaned with a jet of compressed air or renewed as necessary.



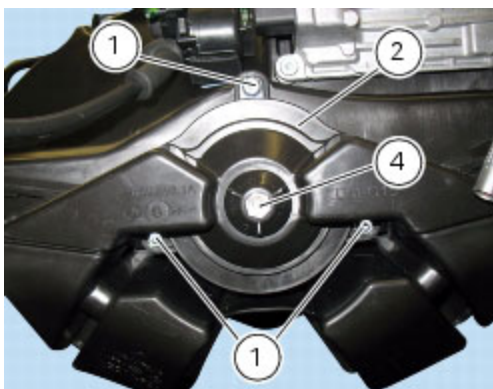
Important

A clogged air filter will reduce air intake and engine power, increase fuel consumption and cause a build up of deposits on the spark plugs. Do not run the engine without the air filter. This will draw foreign material present in the air into the engine and may damage the engine.

Install the cartridge in the airbox cover and refit all the removed parts; tighten the screws (1) and (4) to the specified torque (Sect. C 3, [Frame torque settings](#)).

Important

If the motorcycle is used in very damp or dusty conditions, the air filter cartridge must be changed more frequently.



Changing the brake fluid

Warning

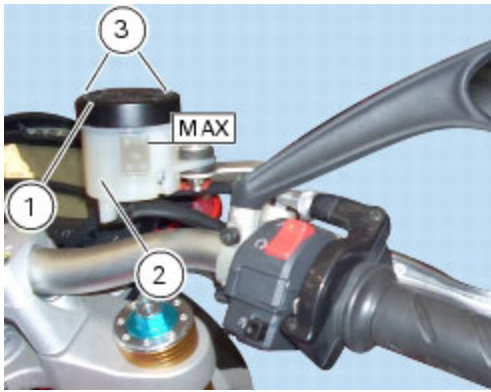
Brake fluid is corrosive and will damage paintwork. Avoid contact with eyes and skin. In the case of accidental contact, wash the affected area thoroughly with plenty of running water.

Changing the fluid in the front brake circuit

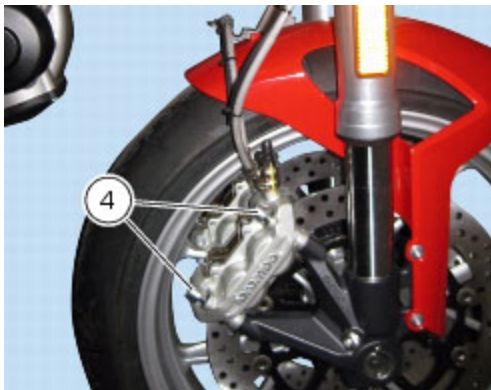
Remove the cover with membrane (1) of the front brake fluid reservoir (2) by unscrewing the screws (3).

Siphon off the fluid from the reservoir (2).

Replace the membrane (without the plug) on the tank to avoid brake fluid splashes during the following operations.



Move the brake callipers pistons back: to carry out this operation unscrew the fixing screws (4) of the callipers on the fork plates and push, by removing them from each other, both pads from every calliper; during this operation be careful with the fluid level, that must always be aspirated each time that it flows back in the tank.



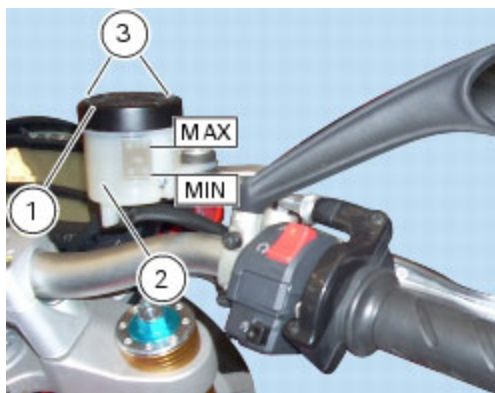
Once that all pistons of both callipers are fully moved back and that all the liquid in the tank has been aspirated, connect to the bleed valve (5) a transparent tube by immersing the end in a container placed on the floor.



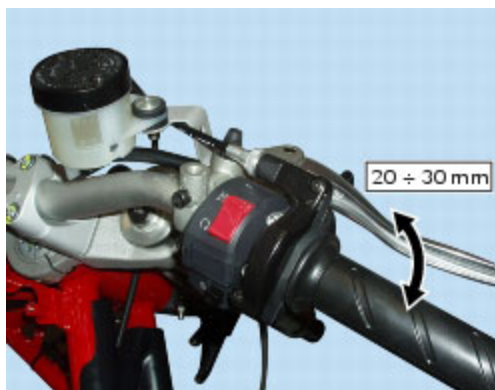
Allow the fluid to flow from the bleed valve (5) until it changes colour.

Tighten the bleed valve to the specified torque (Sect. C 3, [Frame torque settings](#)).

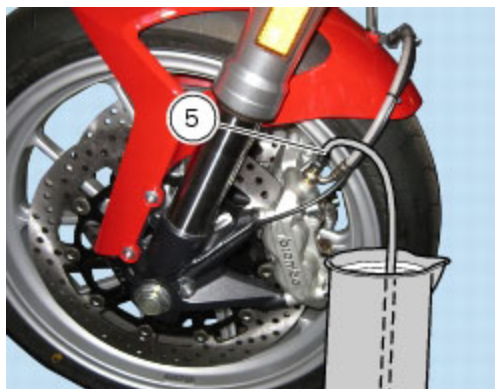
Fill the reservoir (2) with new brake fluid up to the MAX mark.
Refit the cover (1) with membrane and tighten the screws (3).



Operate the lever to reach the 20-30 mm end of stroke and keep the lever in this position by using a non-elastic clamp.



Loosen the bleed valve of the left calliper (from the rider view) (5) and then take the lever through the whole stroke to allow fluid to escape.
The lever is supported to the grip.



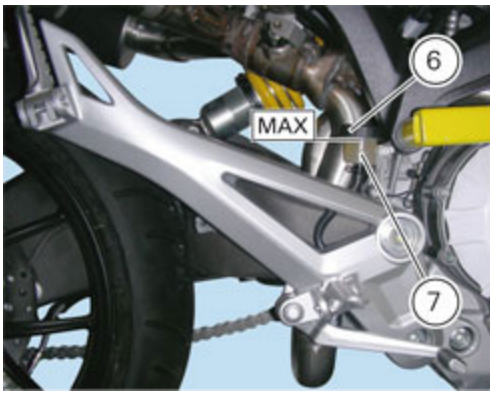
Tighten the bleed valve (5) to the specified tightening torque (Sect. C 3, [Frame torque settings](#)) then release the lever.
Repeat the operation described above until the old liquid flows completely.
Then, with the bleed valve definitely closed to the specified torque actuate repeatedly the lever until a pressure is detected in the brake system.

Warning

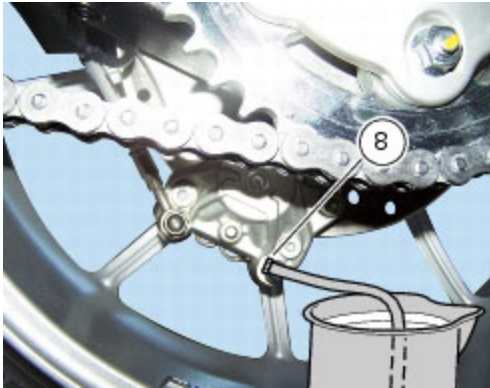
After eliminating the old fluid from the tank, during the refilling operation always keep the oil level above the MIN mark to prevent the formation of air bubbles inside the circuit.

Changing the rear brake circuit fluid

Unscrew the cap (6) of the rear brake fluid reservoir (7).



Attach a length of transparent plastic tubing to the bleed valve (8) and insert the other end of the tubing in a container placed on the floor.

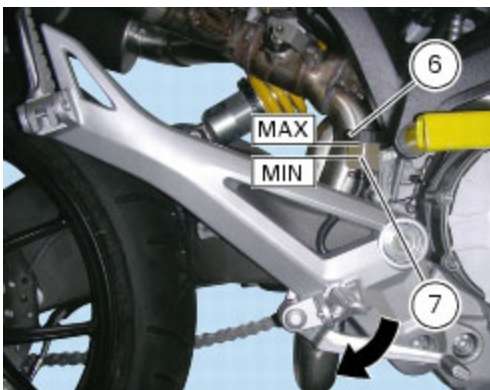


Siphon off the fluid from the reservoir (6).
Fill the reservoir (7) with new brake fluid up to the MAX mark.

Warning

While filling the system, always keep the fluid level above the MIN mark to prevent the formation of air bubbles in the circuit.

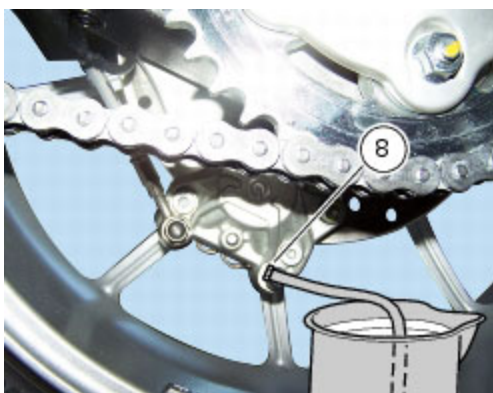
Allow the fluid to flow from the bleed valve (8) until it changes colour.
Operate the brake pedal two or three times to pressurise the circuit.
Hold the pedal pressed down.



Loosen the bleed valve (8) to allow fluid to escape.
The pedal is at the end of the stroke in lower position.
At this point, tighten the bleed valve (8) to the specific tightening torque (Sect. C 3, [Frame torque settings](#)) and release the pedal; press the pedal.
Repeat the operation described above until the old liquid flows completely.

Warning

After eliminating the old fluid from the tank, during the refilling operation always keep the oil level above the MIN mark to prevent the formation of air bubbles inside the circuit.



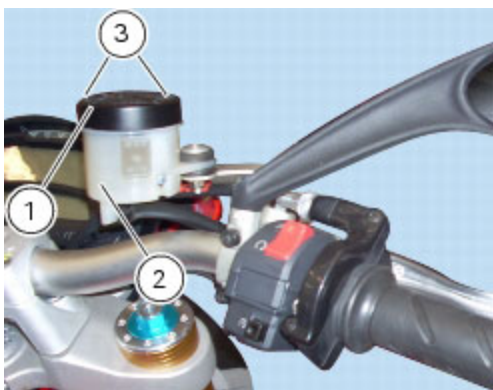
Draining the brake circuit



Warning

Brake fluid is corrosive and will damage paintwork. Avoid contact with eyes and skin. In the case of accidental contact, wash the affected area thoroughly with plenty of running water.

Remove the cover (1) and membrane from the front brake fluid reservoir (2) by loosening screws (3) and the cover (6) from the rear brake fluid reservoir (7).



Connect a commercial brake bleeder to the front brake calliper bleed valve (5) or rear brake calliper bleed valve (8).

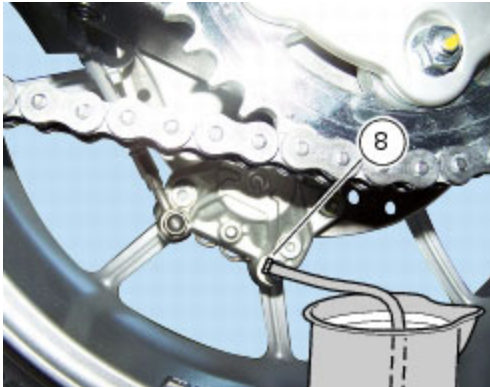
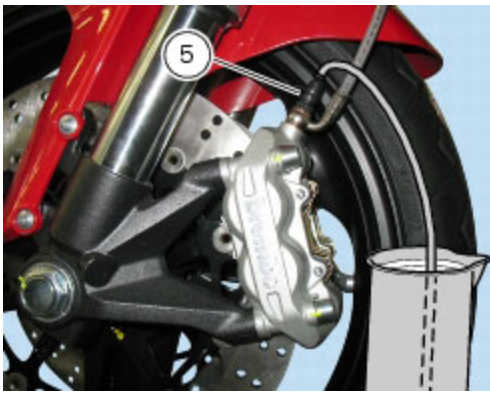


Note

Follow the manufacturer's instructions when using a commercial brake bleeding tool.

Open the bleed valve and pump with the bleeding tool until no more fluid emerges.

If you do not have a brake bleeding tool available, attach a length of transparent plastic tubing to the front calliper bleed valve (5) or rear calliper bleed valve (8) and immerse the other end of the tubing in a container on the floor containing old brake fluid.



Loosen the bleed valve.
Operate the brake lever (or pedal) until all the fluid has drained out.
Repeat this operation for each brake calliper.



Note (front system)

With this procedure in the front system the front callipers fluid-filled remain; in case the callipers also need to be emptied, it is necessary to carry out a retreat piston operation described in the chapter "[Changing the fluid in the front brake circuit](#)" keeping the bleeder (if available) still running and connected to the bleed valve.



Note (rear system)

If there is no bleeder, carry out piston retreat operation described in the chapter "[Changing the fluid in the front brake circuit](#)".

Filling the brake system with fluid



Warning

Brake fluid is corrosive and will damage paintwork. Avoid contact with eyes and skin. In the case of accidental contact, wash the affected area thoroughly with plenty of running water.

Refill the tanks (2) and (7) with specific oil (Sect. C 2, [Fuel, lubricants and other fluids](#)) taken from an intact container.





Important

During the following operation, the fluid level must remain topped over the MIN mark all times. The end of the transparent plastic pipe must remain immersed in the discharged fluid at all times.

Connect the bleed tool to bleed valve (5) or (8).

Operate the lever or the brake pedal and keep it actuated during the refilling.

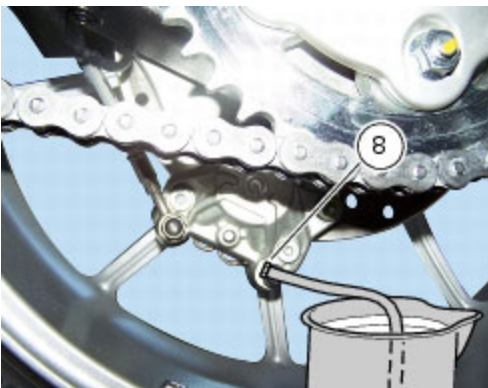
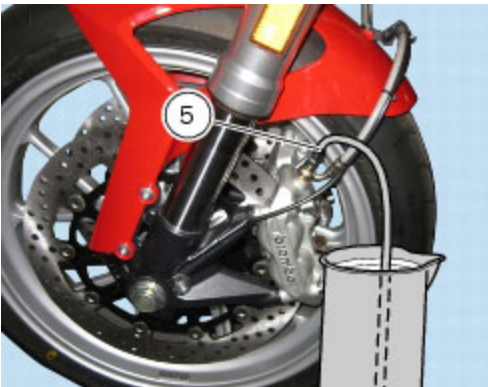
Note

Follow the manufacturer's instructions when using a commercial brake bleeding tool.

Aspirate with bleeder and loosen the bleed valve (5) or (7) always checking that the level of the corresponding tank does not fall below the MIN.

Continue this procedure until all the air is expelled from the system.

Tighten the bleed valve to the specified tightening torque (Sect. C 3, [Frame torque settings](#)).



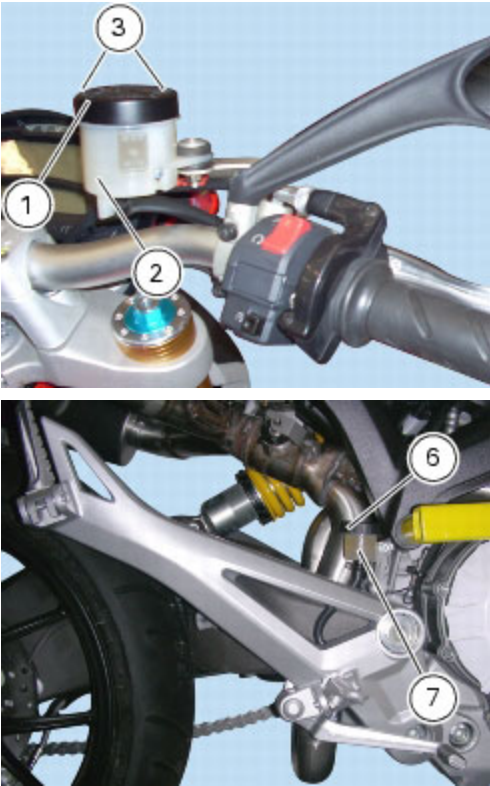
If you do not have a brake bleeder available, connect a length of transparent plastic tubing to the bleed valve as outlined in the draining procedure.

Press the lever or the pedal half way or until the pressure in the system is detected, open the bleed valve and take the lever or pedal through the whole stroke; tighten the bleed valve to the specified torque (Sect. C 3, [Frame torque settings](#)) and release the lever or the pedal.

Important

Do not release the brake lever (or pedal) until the bleed valve has been fully tightened.

Finally check that with the bleed valves closed the activation of the pressure with the lever or pedal is correctly. Tighten to the prescribed tightening torque (Sect. C 3, [Frame torque settings](#)) all the bleed valves and install the protection cover. Level the liquid and refit the cover (1) by tightening the screws (3) on the tank (2) or the cap (6) on the tank (7).



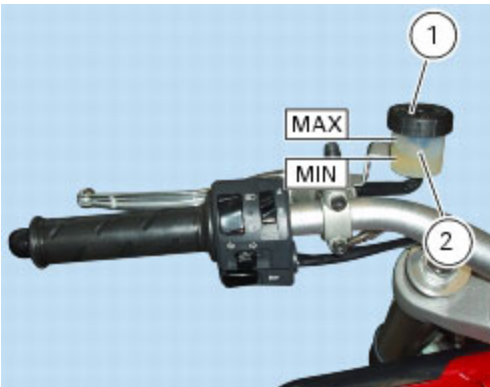
Changing the clutch fluid



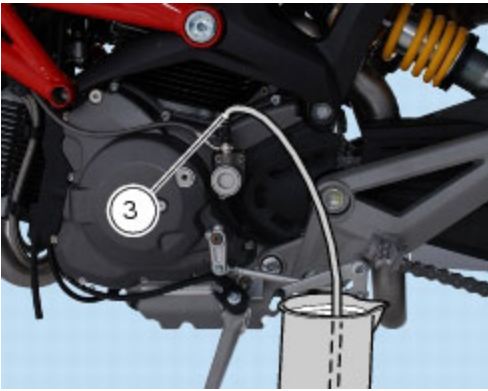
Warning

Clutch fluid will damage painted surfaces if spilled on them. It is also very harmful if it comes into contact with the skin or with the eyes; in the event of accidental contact wash the affected area with abundant running water.

Loosen the cover with membrane (1) of the clutch fluid reservoir (2).



Attach a length of transparent plastic tubing to the bleed valve (3) and insert the other end of the tubing in a container placed on the floor.



Operate the clutch lever two or three times until the circuit is pressurised.
 Hold the lever pulled in towards the grip.
 Open the bleed valve (3) to allow fluid to escape.
 Open the bleed valve (3) to allow fluid to escape.



Warning

During the filling operation, always keep the fluid level above the **MIN** mark to prevent the formation of air bubbles in the circuit.

Operate the clutch lever two or three times until the circuit is pressurised.
 Hold the lever pulled in towards the grip.
 Allow the fluid to flow from the bleed valve (3) until it changes colour. Retighten the bleed valve (3) and tighten to the specified torque (Sect. C 3, [Frame torque settings](#)) and restore the correct level of oil in the tank.

Draining the clutch hydraulic circuit



Warning

Clutch fluid will damage painted surfaces if spilled on them. It is also very harmful if it comes into contact with the skin or with the eyes; In the case of accidental contact, wash the affected area thoroughly with plenty of running water.

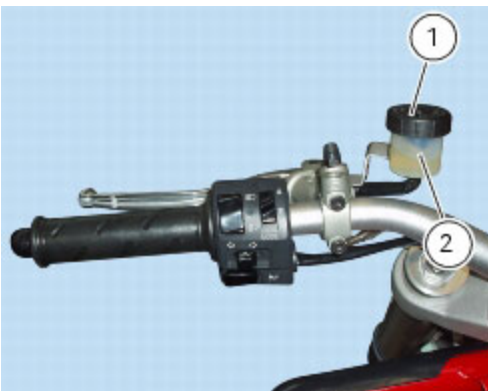
Remove the dust cap to expose the bleed valve (3).
 Connect a clutch circuit bleeding tool to the transmission unit bleed valve (3).



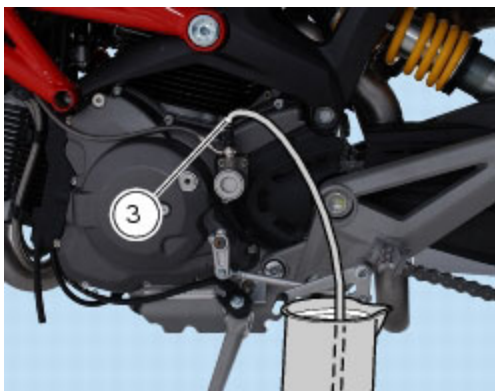
Note

Follow the manufacturer's instructions when using a commercial clutch bleeding tool.

Loosen the cap (1) and membrane from the clutch fluid reservoir (2).

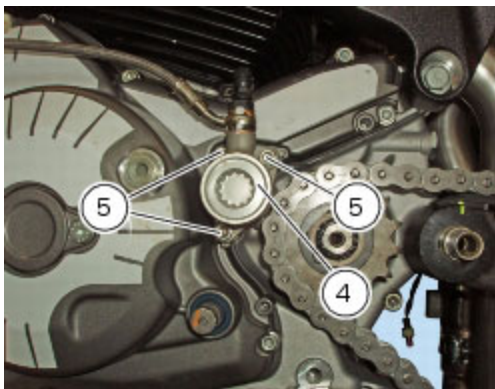


Open the bleed valve and pump with the bleeding tool until no more fluid emerges.
 If you do not have a bleeding tool available, attach a length of transparent plastic tubing to the bleed valve (3) and insert the other end of the tubing in a container of old clutch fluid placed on the floor.
 Unscrew the bleed valve by **1/4** of a turn.



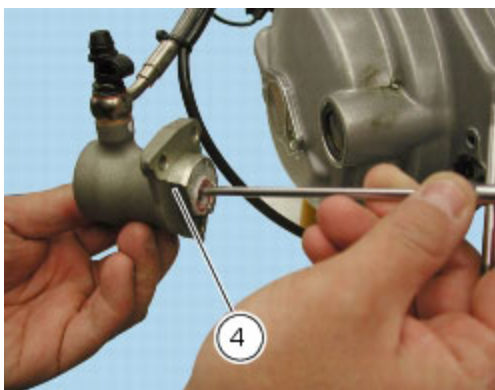
Operate the clutch lever until all the fluid has been expelled.

To help drain the circuit fully, unscrew the three retaining screws (5) and remove the slave cylinder cap from the engine.

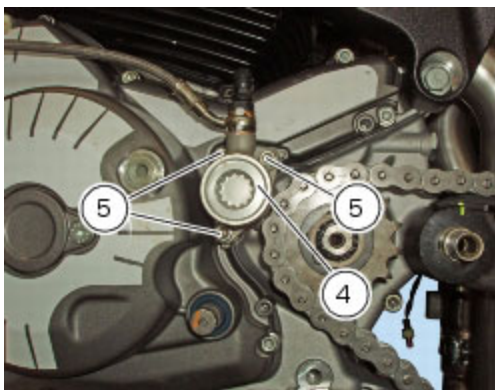


Withdraw the clutch slave cylinder assembly (4).

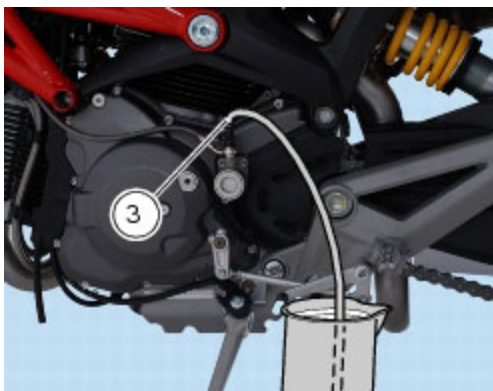
Push in the internal piston to force out all the fluid from inside the cap.



Proceed with the refitting of the cap by tightening the fixing screws (5) to the specified torque (Sect. C 3, [Frame torque settings](#)).



Tighten the bleed valve (3) to the specified torque (Sect. C 3, [Frame torque settings](#)).



Filling the clutch circuit



Warning

Clutch fluid will damage painted surfaces if spilled on them. It is also very harmful if it comes into contact with the skin or with the eyes; In the case of accidental contact, wash the affected area thoroughly with plenty of running water.

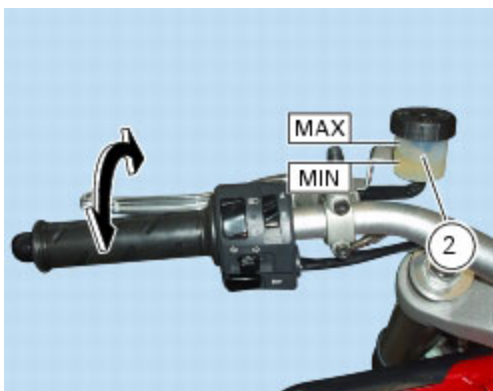
Fill the tank (2) with specified oil (Sect. C 2, [Fuel, lubricants and other fluids](#)) taken from an intact container.



Important

During the following operation, the fluid level must remain topped up at all times. The end of the transparent plastic tubing must remain immersed in the discharged fluid at all times.

Operate the clutch lever several times to fill the circuit and expel any air.



Connect the bleed tool to the bleed valve (3).

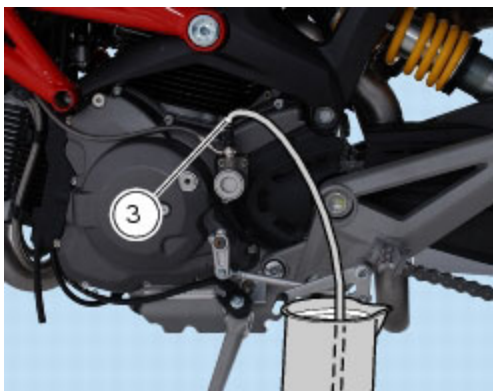


Note

Follow the manufacturer's instructions when using a commercial clutch bleeding tool.

Pump with the bleeder and open the bleed valve (3) making sure that the level does not fall below the **MIN** mark. Repeat the bleeding operation until the fluid emerging from the plastic tube connected to the bleed valve (3) is free of air bubbles.

If you do not have a bleeding tool available, connect a length of transparent plastic tubing to the bleed valve (3) as outlined in the draining procedure.



Open the bleed valve (3) by **1/4** of a turn and operate the clutch lever several times until the fluid starts to emerge from the bleed valve (3).

Pull the lever fully in and then loosen the bleed valve by at least **1/4** turn.

Wait for a few seconds; then release the lever gradually while simultaneously closing the bleed valve (3).



Important

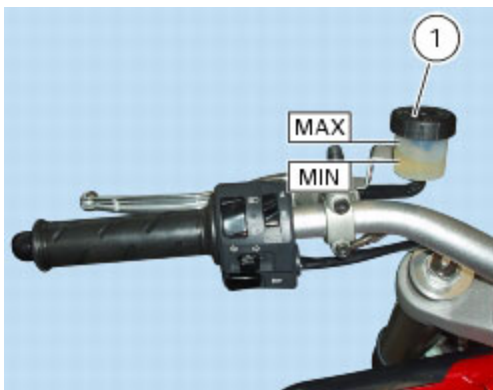
Do not release the clutch lever until the bleed valve has been fully tightened.

Repeat the bleeding operation until the fluid emerging from the plastic tube is free of air bubbles.

Tighten the bleed valve (3) to the specified torque (Sect. C 3, [Frame torque settings](#)) and install the protection cover.

Fill up to **3** mm above the MIN level mark on the reservoir.

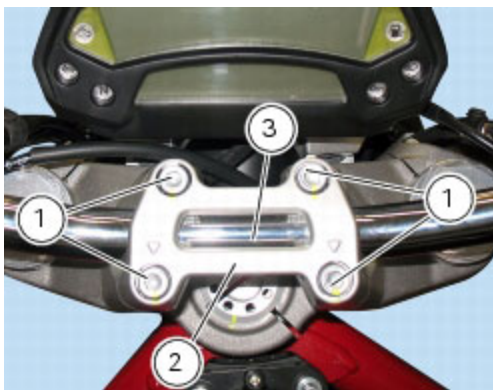
Tighten the cover (1) with the membrane.



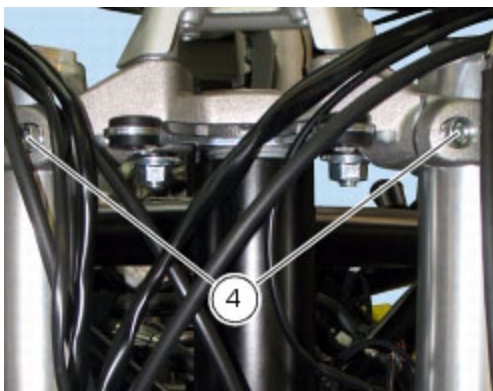
Adjusting the steering head bearings

Excessive handlebar play or the forks shaking in the steering head indicate that the play in the steering head bearings requires adjustment. Proceed as follows.

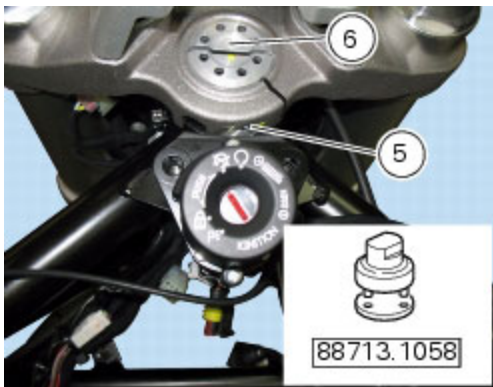
Unscrew the four screws (1), remove the two clamps (2) and remove the handlebar (3) from the steering head, positioning it out of the way so it does not hinder subsequent operations.



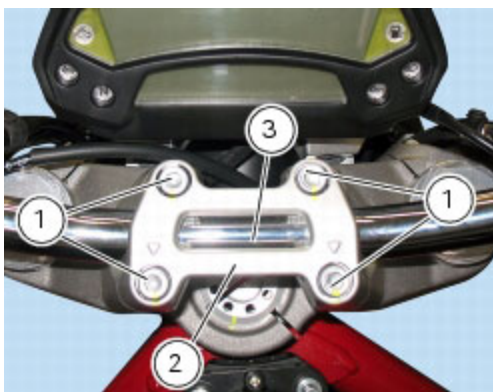
Slacken the clamp screws (4) securing the steering head to both fork legs.



Loosen the clamp screw (5) that holds the steering tube to the steering head.
Using service tool no. **88713.1058**, turn the nut (6) to the specified torque (Sect. C 3, [Frame torque settings](#)).

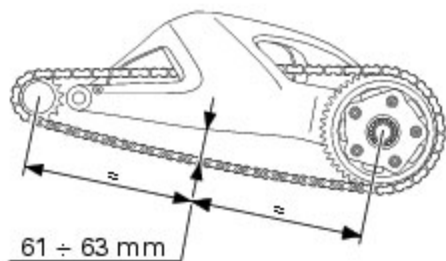


Tighten all previously loosened screws to the specified tightening torque (Sect. C 3, [Frame torque settings](#)).
Reposition handlebar (3), install U-bolt (2) and tighten the four screws (1) to the specified torque (Sect. C 3, [Frame torque settings](#)).



Adjusting the chain tension

Turn the rear wheel slowly to find the position at which the chain is at its most taut.
With the motorcycle on its side stand, press with a finger in the centre of the bottom run of the chain and measure the distance between the centres of chain link pins and the aluminium swingarm. The distance should be within **61** to **63** mm.



To adjust the tension, proceed as follows.

Slacken off the two clamp screws (1) that secure the rear wheel hub to the swingarm.

Fit the hook spanner code **88713.1038** inserting its tooth in the eccentric hub (2).

Rotate the eccentric hub (2) to obtain the correct chain tension.

Turn anticlockwise to tighten the chain; clockwise to loosen (from chain side).

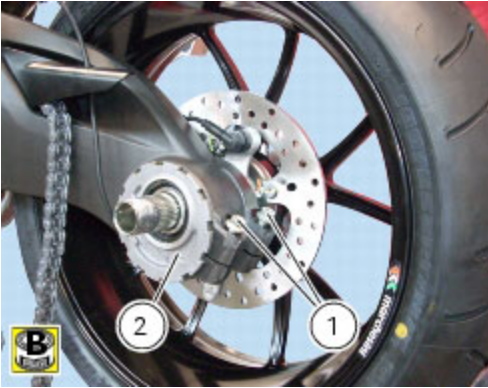
Important

An incorrectly tensioned chain will lead to accelerated wear of the transmission components.

If the screws (1) are removed, lubricate with specified grease underside and thread, then tighten the screws (1) to the specified tightening torque proceeding with sequence 1-2-1.

Warning

The correct tightening of the fixing screws of the eccentric hub is essential for the safety of the rider and the passenger (Sect. C 3, [Frame torque settings](#)).



Checking brake pad wear and changing brake pads

Warning

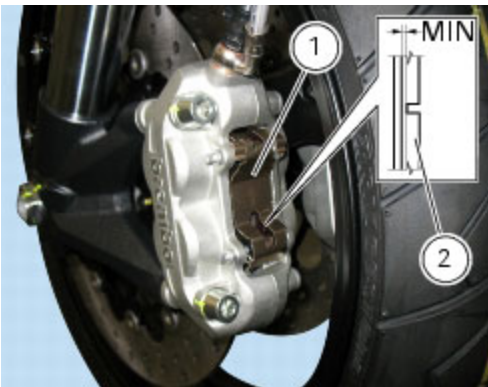
Brake fluid is corrosive and will damage paintwork. Avoid contact with eyes and skin. In the case of accidental contact, wash the affected area thoroughly with plenty of running water.

Important

On handing over the motorcycle after changing the brake pads, inform the customer that the front brake must be used gently for the first **100** km to allow the pads to bed in completely.

Checking front brake pad wear

Look through the slot in the calliper clip (1) to make sure the groove in the brake pad friction material (2) is still visible.



Important

If either of the pads is worn, both pads must be renewed as a set.

Change the brake pads as follows.

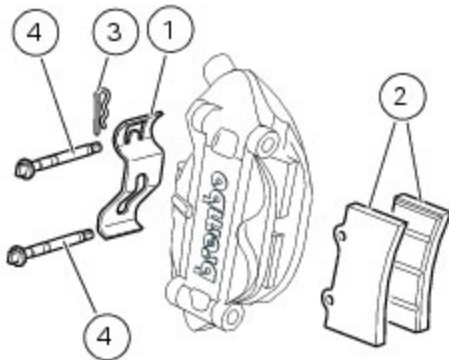
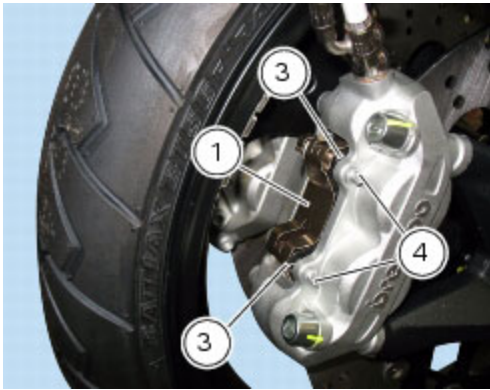
Remove the cotter pin (3) from the pad retaining pins.
Withdraw the pad retaining pins (4) from the outside.
Remove the clip (1) from between the two calliper halves.
Force the calliper pistons back into their bores by forcing the old brake pads apart.
Remove the worn pads (2).



Note

Change pads that have a shiny or “vitrified” appearance.

Insert the new pads and clip (1). Slide in the pad retaining pin (4) and secure it in position with the cotter pin (3).



Operate the brake lever repeatedly so that the pads are firmly bedded in against the disc by the force of the brake fluid.

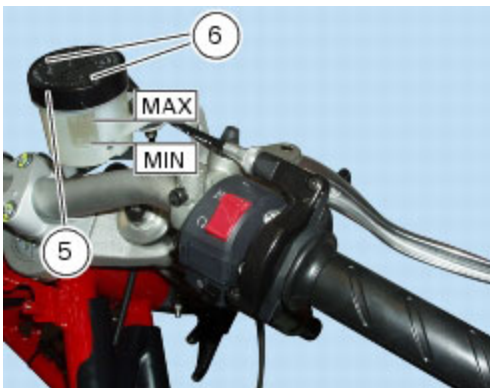
Check that the level in the cylinder tank is not below the MIN mark. If necessary, top up as follows. Turn the handlebar so that the reservoir is level.

Unscrew the two crosshead screws (6) and remove the reservoir cover (5).

Remove the inner membrane from the reservoir.

Top up to the maximum level with the specified brake fluid (Sect. C 2, [Fuel, lubricants and other fluids](#)).

Refit all previously removed components.



Checking rear brake pad wear

Check through the slot between the two half callipers that there is at least **1** mm of friction material on the pads.

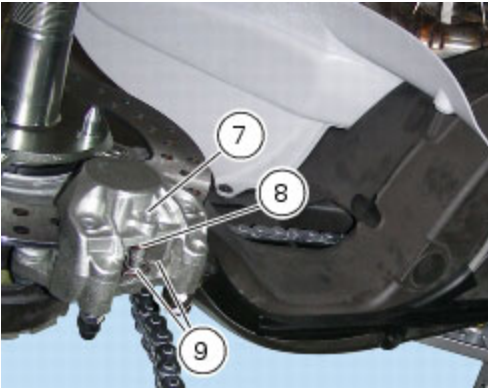
● Important

If either of the pads is worn, both pads must be renewed as a set.

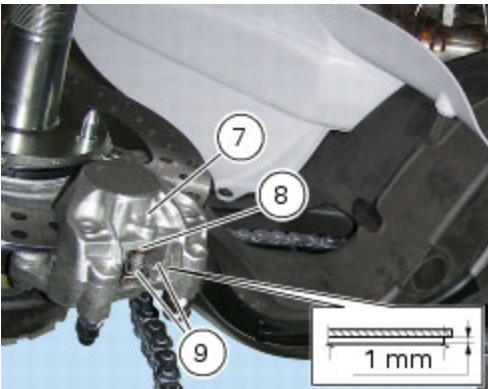
Change the brake pads as follows.

Remove the snap ring (7) (on the inner side of the calliper) from the pad retaining pin. Force the brake pads apart to push the calliper pistons into their housings. Withdraw the brake pad retaining pin (8) and pull it out.

Remove the clip (9) from between the two calliper halves. Remove the worn pads.



Insert the new pads and clip (9). Slide in the pad retaining pin (8) and secure it in position with the snap ring (7).

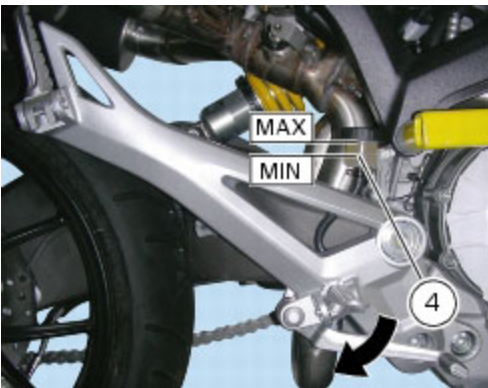


👁 Note

Change pads that have a shiny or “vitrified” appearance.

Operate the brake pedal repeatedly so that the pads are bedded in against the disc by the force of the brake fluid.

Check that the brake fluid level in the reservoir (4) is between the MIN and MAX marks. If not, unscrew the reservoir cover (4) and top up with the recommended brake fluid.



👁 Note

If it proves difficult to change the brake pads, first remove the callipers from the motorcycle (Sect. G 6, [Removal of the rear brake calliper](#)).

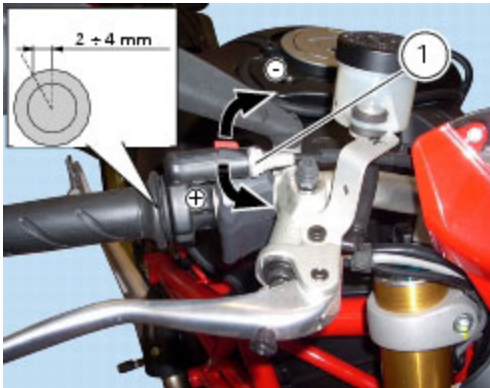
⚠ Warning

Being the brake callipers a security component of the vehicle, follow instructions indicated in Sect. G 3, [Removal of the front brake system](#) and Sect. G 6, [Removal of the rear brake calliper](#), and pay attention, during the refitting, to the tightening torque (Sect. C 3, [Frame torque settings](#)) of the fixing screws of the callipers.

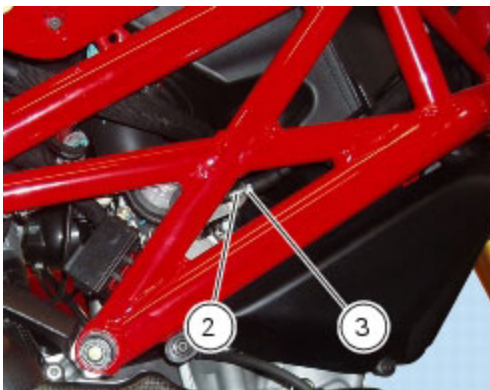
Adjusting the throttle cable

The throttle grip in all steering positions must have free play, measured on the periphery of the flange of the grip, of **2 ÷ 4 mm**.

If necessary, adjust the free play with the adjuster (1) located on the twistgrip.



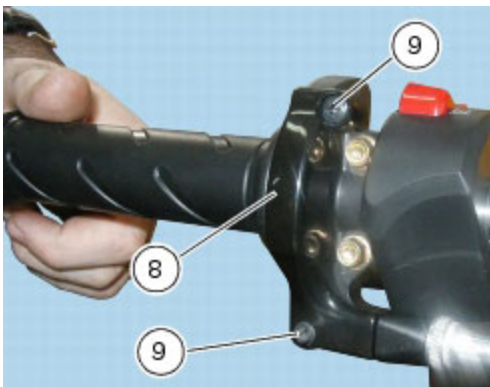
Major adjustments of the throttle cable are possible by working adjuster (2) located on throttle body right-hand side. Remove the gaiter, loosen the lock nut (3) and work adjuster (2) to set the required play. Tighten the lock nut and refit the gaiter.



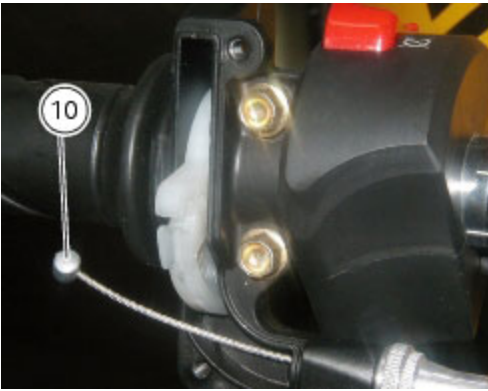
The condition of the throttle cable and choke cable sheaths should be checked at regular intervals. The plastic covering should show no signs of pinching or cracking.

Lubricate the ends of the throttle control cables with the prescribed grease periodically to ensure they run freely. Operate the control to check that the inner cable slides smoothly inside the outer cable: if you feel excessive resistance or stiffness, renew the cable.

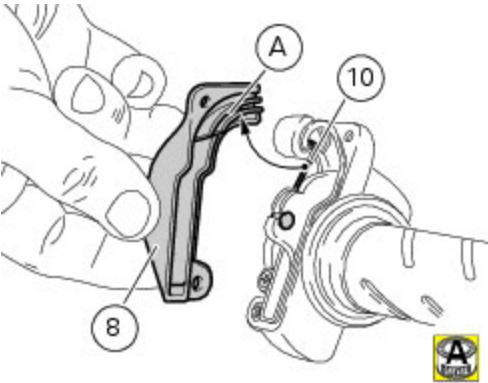
To lubricate throttle cable, undo the two screws (9) and remove the upper half of the twistgrip housing (8).



Grease the end of the cable (10) and the race.

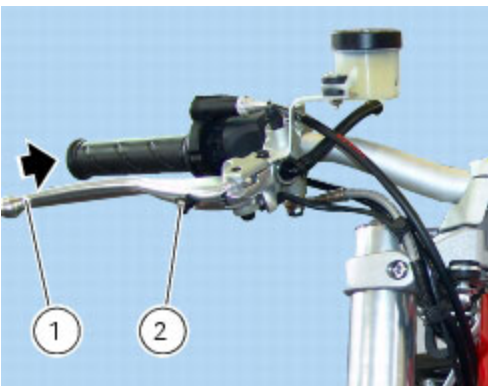


Carefully close the throttle assembly, inserting the cable (10) in the guide (A) on the upper twistgrip housing (8). Replace and tighten the two screws (9).



Adjusting the clutch lever and front brake lever

Lever (1) disengages the clutch. It features a dial adjuster (2) for lever distance from the twistgrip on handlebar. The lever distance can be adjusted through 10 clicks of the dial (2). Turn clockwise to increase lever distance from the twistgrip. Turn the adjuster counter clockwise to decrease lever distance. When the clutch lever (1) is operated, drive from the engine to the gearbox and the drive wheel is disengaged. Correct use of the clutch lever is very important in all riding situations, especially when moving off. The position of the front brake lever can be adjusted in the same way.





Warning

Clutch lever and front brake lever shall be adjusted when motorcycle is at a standstill.



Important

Using the clutch properly will avoid damage to transmission parts and spare the engine.



Note

The engine can be started with the side stand down and the gearbox in neutral. If starting with a gear engaged, pull in the clutch lever (in this case the side stand must be up).

Adjusting the position of the gear change and rear brake pedals

The position of the gear change and rear brake pedals in relation to the footrests can be adjusted to suit the preferred riding position.

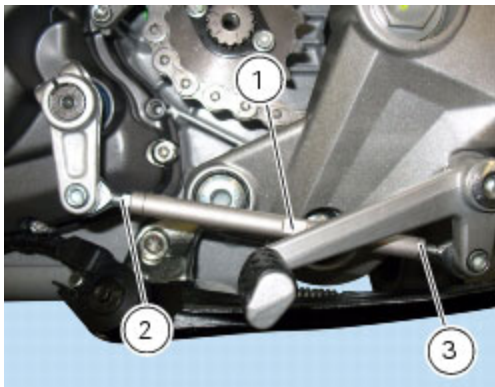
To adjust the position of the gearchange pedal, proceed as follows:
hold the linkage (1) and slacken the lock nuts (2) and (3).



Note

The lock nut (2) has a left-hand thread.

Turn the tie-rod (1) using an open-ended wrench on the flats to move the gear change pedal to the required position.



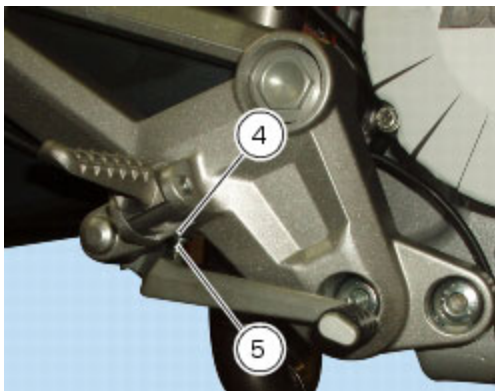
Tighten both lock nuts onto the rod.

To adjust the position of the rear brake pedal, proceed as follows.

Loosen lock nut (4).

Turn the pedal travel adjustment screw (5) until the pedal is in the desired position.

Tighten the lock nut (4).



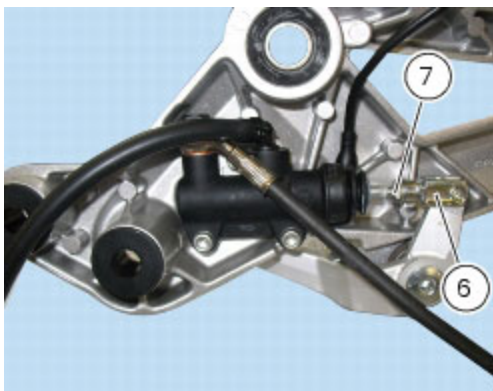
Verify, by acting manually on the pedal, that there is a clearance of about **1.5 ÷ 2 mm** before starting the braking operation.

If not, check to modify the length of the cylinder push-rod in the following mode.

Slacken off the lock nut (7) on the pushrod.

Screw the rod into the fork (6) to increase play, or unscrew it to reduce play.

Tighten the lock nut (7) and recheck the pedal free play.



Adjusting the front fork

The front fork can be adjusted for rebound, compression and spring preload.

The settings are adjusted by way of external adjuster screws:

- 1) rebound damping;
- 2) inner spring preload;
- 3) compression damping.

Park the motorcycle in a stable position, upright.

Turn the adjuster (1) on fork leg top with a flat screwdriver (1100 ABS) or with the relevant tool (1100 S ABS) to adjust rebound damping.

Turn the adjuster (3) on the bottom of each fork leg with a flat-blade screwdriver to adjust the compression damping setting.

As you turn the adjusters (1) and (3), you will hear them click. Each click corresponds to different damping setting. The stiffest damping setting is obtained with the adjuster turned fully clockwise to the "0" position. Start with this position and turn anticlockwise. Count the clicks, which correspond to position 1, 2 and so forth.

To change the spring preload for each fork leg, turn the adjuster (2) with a 22 mm hex wrench.

The STANDARD factory settings are as follows:

1100 ABS

compression: 1.5 turns $\pm 1/4$ turn;

rebound: 1.5 turns $\pm 1/4$ turn.

spring preload (A): 10 mm (3 turns from fully open position).

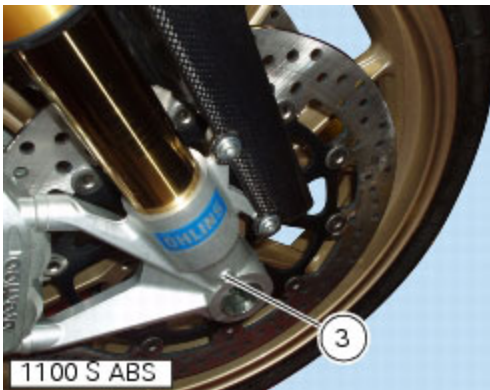
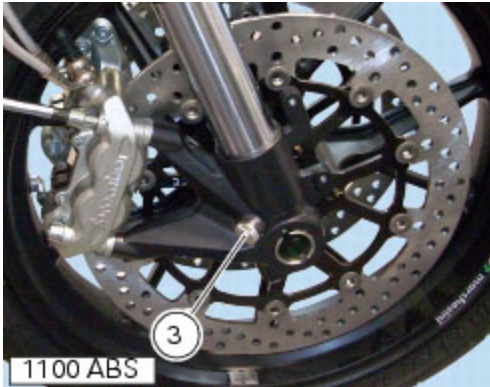
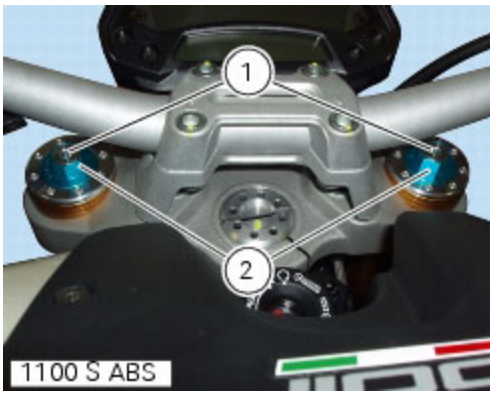
1100 S ABS

compression: 8 clicks.

rebound: 12 clicks.

spring preload (A): 10 mm (3 turns from fully open position).





Adjusting the rear shock absorber

The shock absorber has external adjusters that enable you to adjust the suspension to suit the motorcycle set up. The adjuster (1) located on the left-hand side, where the upper end of the shock absorber is attached to the rear subframe, controls rebound damping.

Turn the adjuster (1) clockwise to increase damping **H**; or anticlockwise to reduce damping **S**.

STANDARD setting from the fully closed position (clockwise):

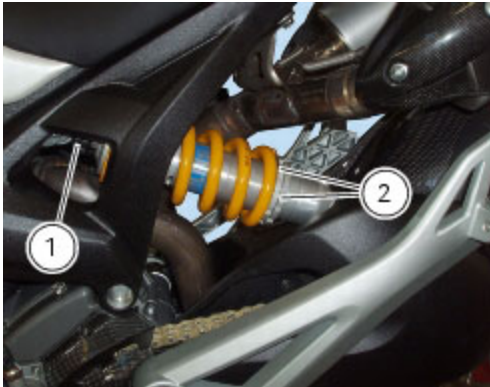
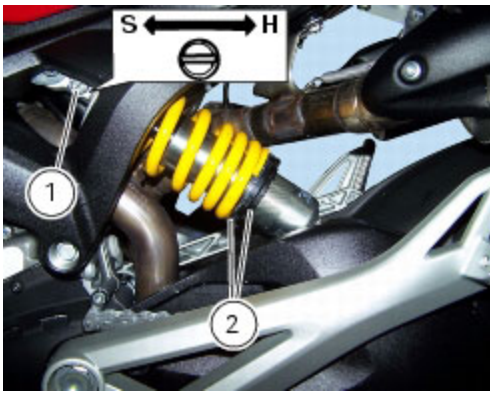
1100 ABS

- loosen adjuster (1) by 13 clicks.
Spring preload: 18 mm.

1100 S ABS

- loosen adjuster (1) by 15 clicks.
Spring preload: 10 mm

The two nuts (2) on the upper part of the shock absorber serve to adjust the preload on the external spring. To change spring preload, slacken off the upper ring nut. Then tighten or slacken the lower ring nut to increase or decrease spring preload.



Warning

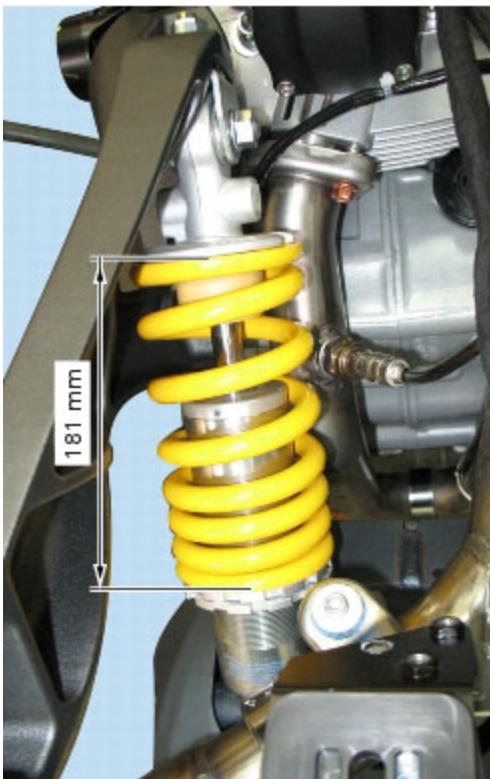
To turn the preload adjuster nuts, use only a specific tool and take care, otherwise the wrench pins may slip off the nut. This could injure your hands. Never use a wrench with a too small pin or with a too short handle.



Important

When adjusting the spring preload, do not exceed a spring length of **181** mm to avoid damaging the swingarm.

Standard length of the preloaded spring on the shock absorber with the rear wheel raised off the ground: **150±1** mm.





Warning

The shock absorber is filled with high-pressure gas and can cause injuries if inexpertly dismantled.



Important

If the motorcycle is to be ridden with a pillion rider and luggage, we recommend setting the rear shock absorber spring preload to the maximum to ensure the best handling and proper ground clearance at all times. It may also be necessary to adjust the rebound damping accordingly.

Adjusting the exhaust valve bowden cable



Note

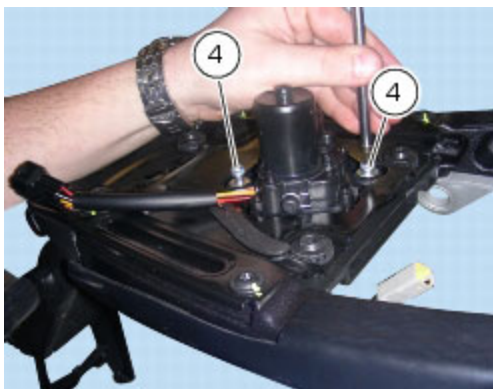
Adjustment procedure for all models with one control cable.

Operation	Section reference
Remove the seat	E 3, Removal of the seat

Remove the protection (1), unscrewing the screw (2) and recovering the washers (3) and corresponding spacer.



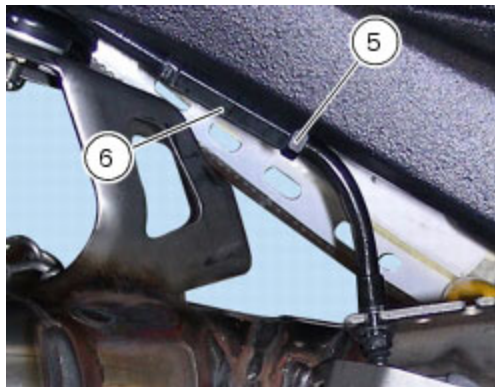
Unscrew the bolts (4) to raise the exhaust valve motor from the rear number plate holder.



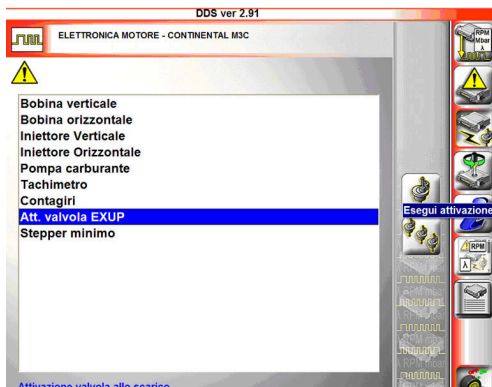
For any position assumed by the vertex of the hexagon (D) flanked by the two cable end fittings, between the vertical position (A) and the position corresponding to the notch (B).



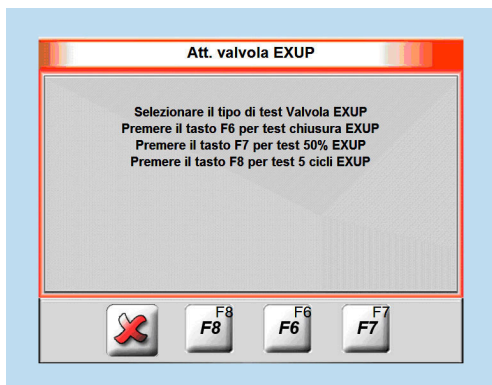
Loosen lock nut (5) from adjuster and tighten the adjuster ring nut (6) at least by 4 turns.



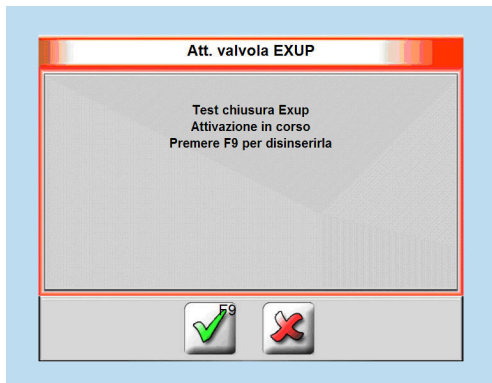
Connect the DDS (Sect. D 5, [Tester power supply](#)) and open the procedure "EXUP valve act.".



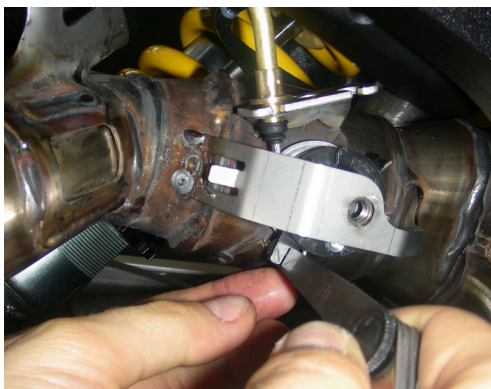
The page opens and you can select to activate the procedure.



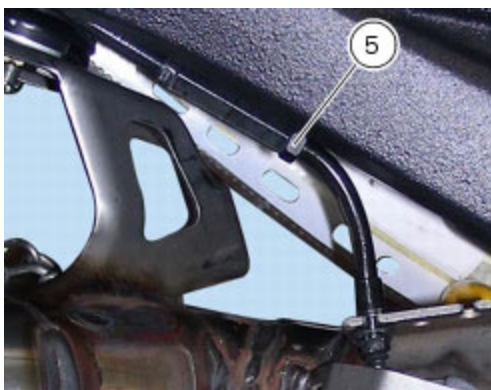
Press "F6" to close the valve that stays closed.



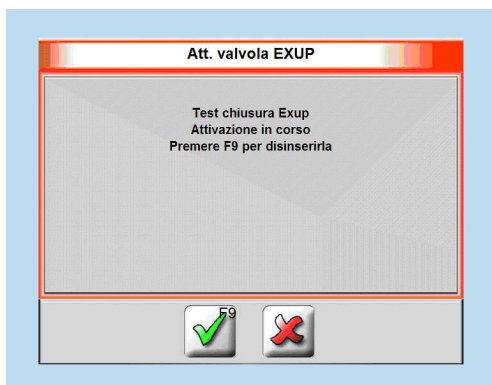
Undo the adjuster of the cable until you obtain a free play in the range of 0.5 mm to 1 mm measured between valve stopper and pulley stop.



Tighten the lock nut (5) to the specified torque (Sect. C 3, [Frame torque settings](#)).

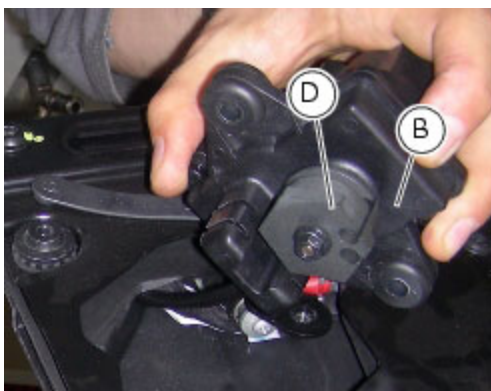


Press button "X" from this DDS page: the activation procedure page closes and system quits without going through test selection.



Disconnect the DDS.

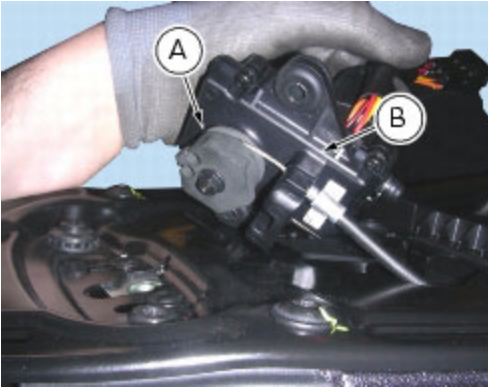
Turn the key to on, let make one complete turn and make sure that the vertex of the hexagon (D) in-between the cable mounting holes is secure/fixed next to the notch (B).



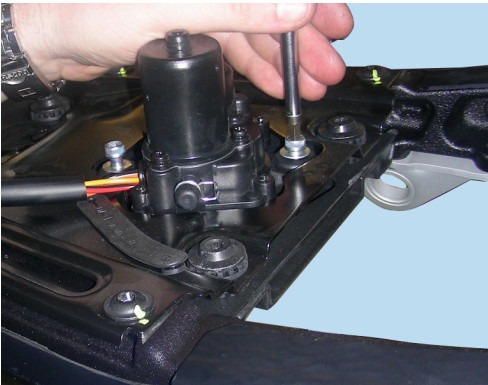
Fitting a new motor and new cable

Insert cable end coming out of the straight sheath length on motor pulley into hole (A) shown.
Insert sheath end into the relevant seat (B) on the motor.

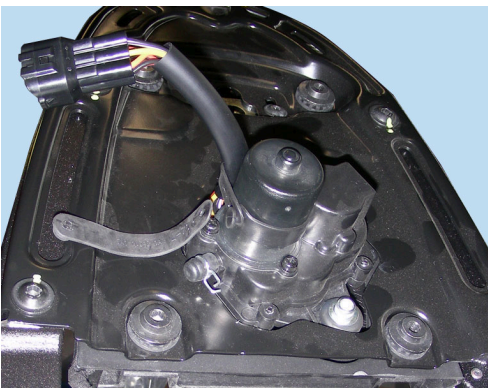
Fit the clip to sheath terminal and make sure that it prevents cable from slipping out.



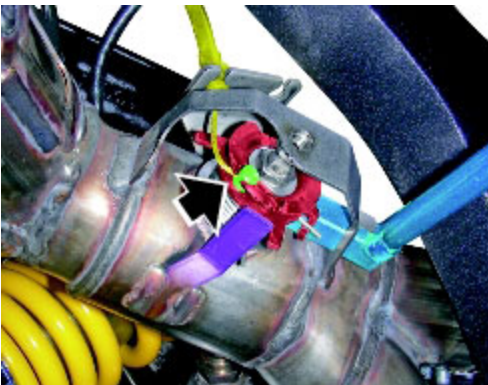
Turn the clip indicated by the arrow to the position shown in the figure.
Route cable in the tunnel provided between plate and number plate holder.
Position and fasten the valve motor to the rear subframe plate by means of two screws and two washers.
Tighten the screws to the specified torque (Sect. C 3, [Frame torque settings](#)).

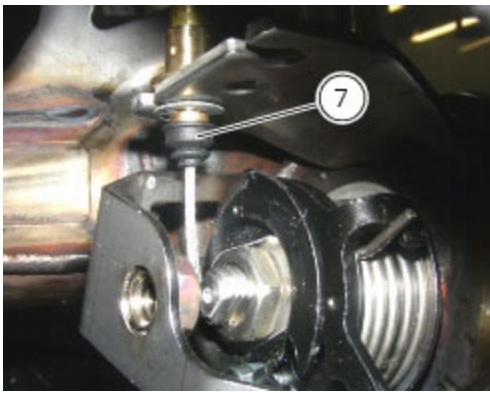


Fasten the electric cable to the motor central body using a rubber tie L82.
Ensure the cable is not touching the metal plate edge (area indicated by the arrow).

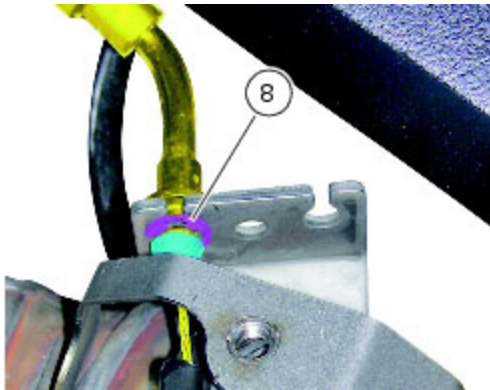


Turn exhaust valve pulley clockwise fully and hold it in this position.
Slide off the dust gaiter (7) from the sheath end.
Fit sheath end into the left-hand hole available on the plate above the valve pulley and refit the dust gaiter.
Fit cable end into the pulley hole indicated in the figure and release pulley so as cable is taut again.

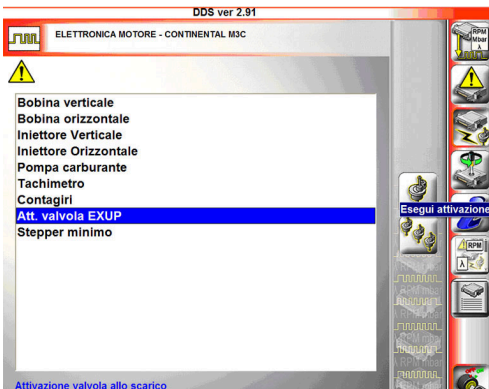




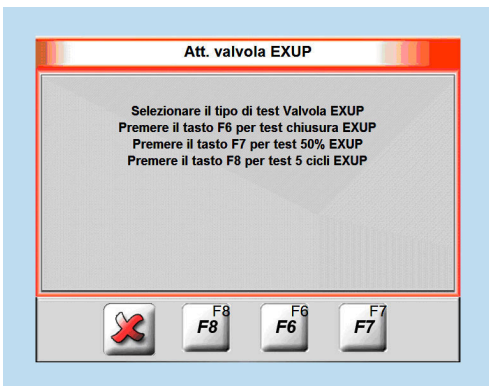
Lock cable end to plate by means of the circlip (8).



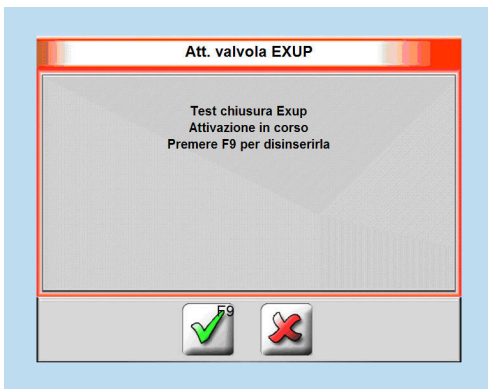
Connect the DDS (Sect. D 5, [Tester power supply](#)) and open the procedure "EXUP valve act.".



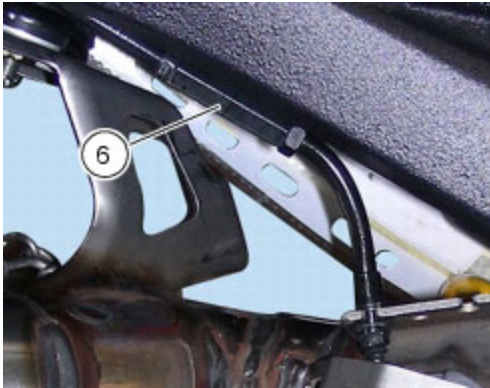
The page opens and you can select to activate the procedure.



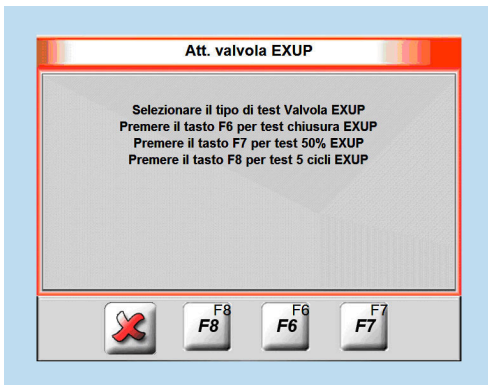
Press "F6" to close the valve that stays closed.



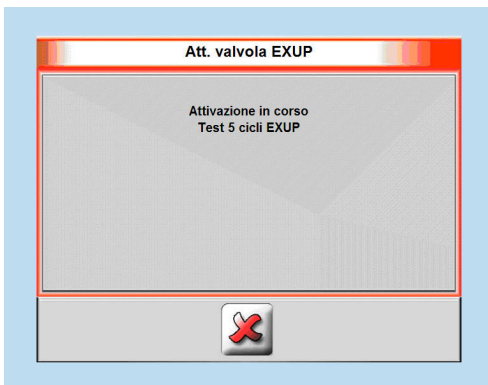
Loosen adjuster ring nut (6) on cable until pulley gets against the limit stop, further loosen it by 1.5 turns.



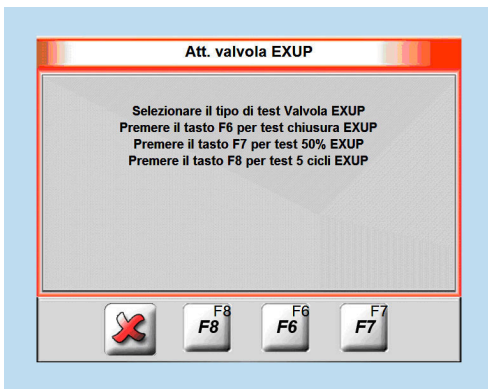
On the DDS, press "F9": to go back to test selection page.



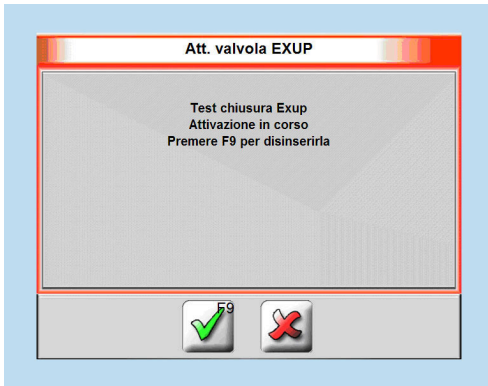
Press "F8" to open function "EXUP 5-cycle test".
The 5-cycle test is performed.



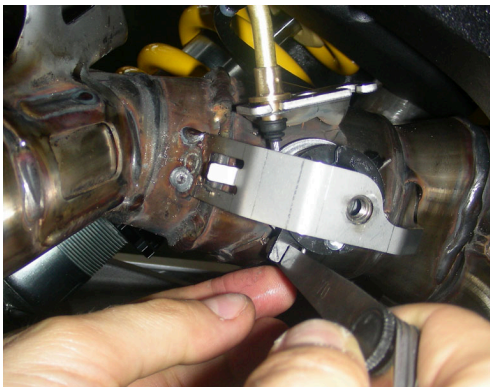
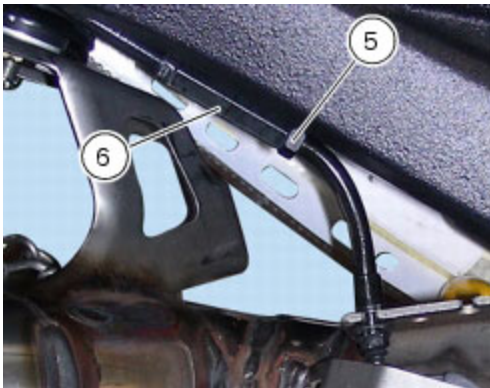
On the DDS, press "F9": to go back to test selection page.



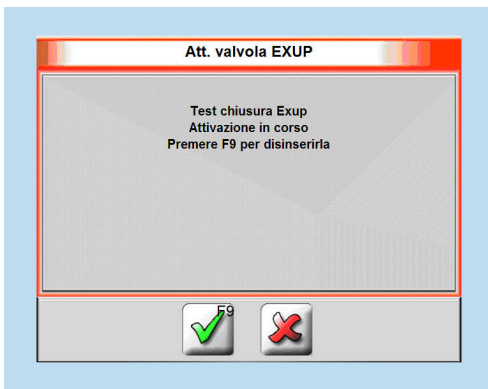
Press "F6" to close the valve that stays closed.



Tighten the adjuster ring nut (6) on cable until you obtain a free play in the range of 0.5 mm to 1 mm measured between limit stop and pulley stop.
Once adjustment is completed, distance between the stop plate welded to the exhaust pipe and the valve pulley front tab must be 0.5 to 1 mm.
Tighten the lock nut (5) to the specified torque (Sect. C 3, [Frame torque settings](#)).



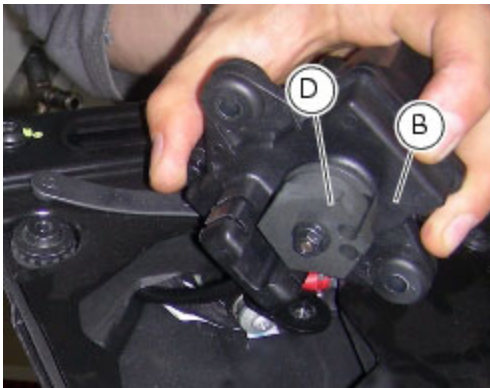
Press button "X" from this DDS page: the activation procedure page closes and system quits without going through test selection.



Disconnect the DDS.

Turn the key to on, let make one complete turn and make sure that the vertex of the hexagon (D) in-between the cable mounting holes is secure/fixed next to the notch (B).

If the test result was positive in the previous step, proceed with the remaining operations. While if result was negative, find failure cause before continuing.



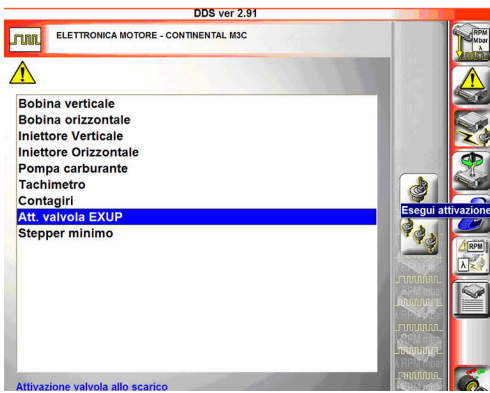
Final checks

Make sure that, upon the following immediate "KEY ON" cycles, the exhaust valve stays in "fully open position", i.e. the valve pulley rear tab must be resting against the stop plate welded to the exhaust pipe.

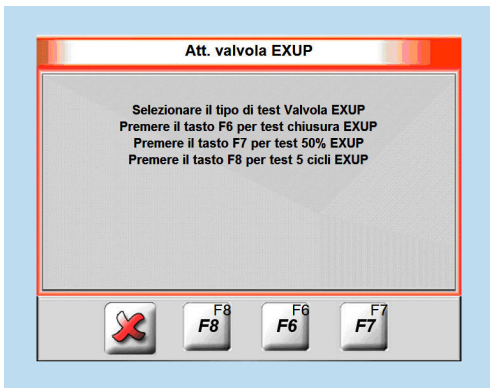


Fitting a new exhaust valve cable with used motor

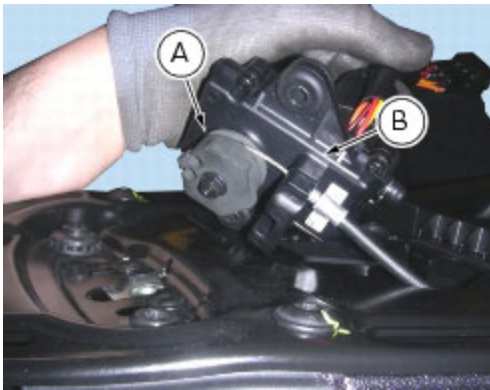
Connect the DDS (Sect. D 5, [Tester power supply](#)) and open the procedure "EXUP valve act.".



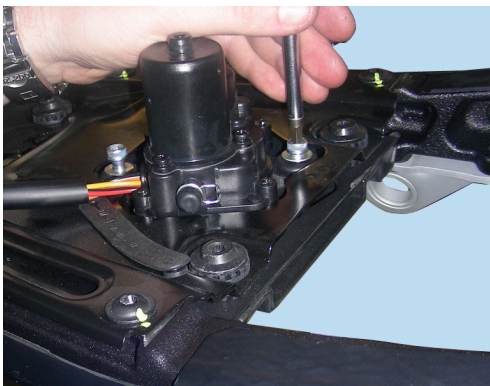
Press "F7" to open function "EXUP 50% test".



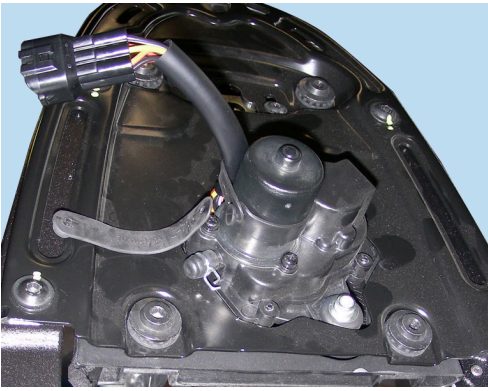
The pulley will set with its vertex (in-between the cable end holes) at the motor centreline.
 Insert cable end coming out of the straight sheath length on motor pulley into hole (A) shown.
 Insert sheath end into the relevant seat (B) on the motor.
 Fit the clip to sheath terminal and make sure that it prevents cable from slipping out.



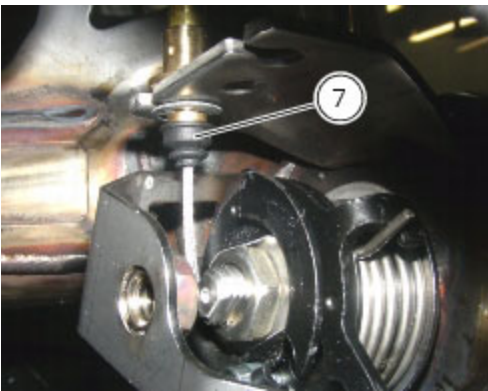
Turn the clip indicated by the arrow to the position shown in the figure.
 Route cable in the tunnel provided between plate and number plate holder.
 Position and fasten the valve motor to the rear subframe plate by means of two screws and two washers.
 Tighten the screws to the specified torque (Sect. C 3, [Frame torque settings](#)).



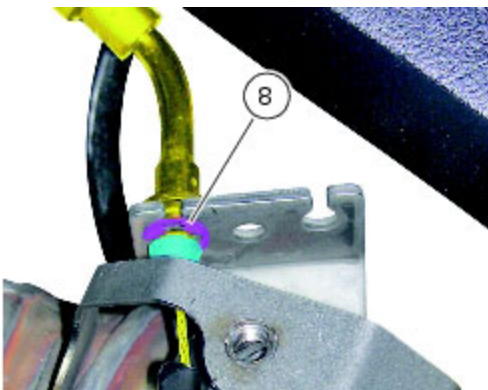
Fasten the electric cable to the motor central body using a rubber tie L82.
 Ensure the cable is not touching the metal plate edge (area indicated by the arrow).



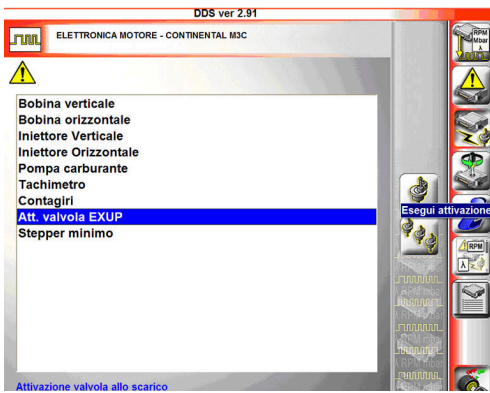
Turn exhaust valve pulley clockwise fully and hold it in this position.
 Slide off the dust gaiter (7) from the sheath end.
 Fit sheath end into the left-hand hole available on the plate above the valve pulley and refit the dust gaiter.
 Fit cable end into the pulley hole indicated in the figure and release pulley so as cable is taut again.



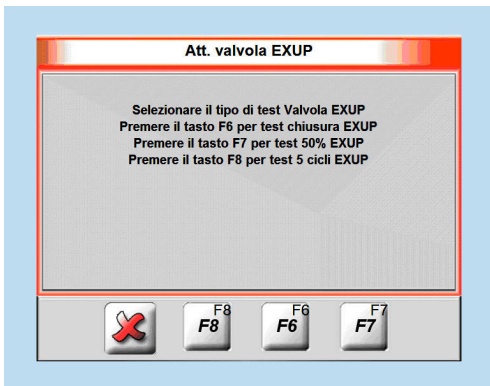
Lock cable end to plate by means of the circlip (8).



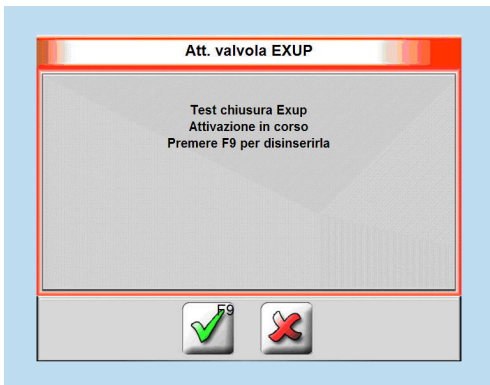
Connect the DDS (Sect. D 5, [Tester power supply](#)) and open the procedure "EXUP valve act.".



The page opens and you can select to activate the procedure.



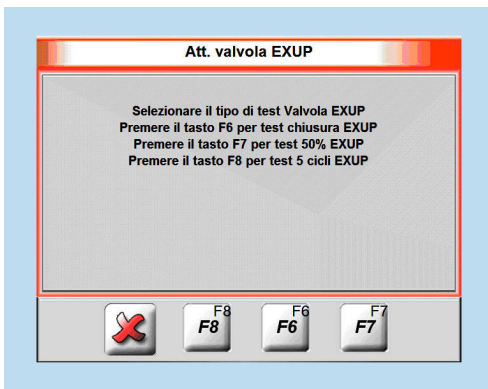
Press "F6" to close the valve that stays closed.



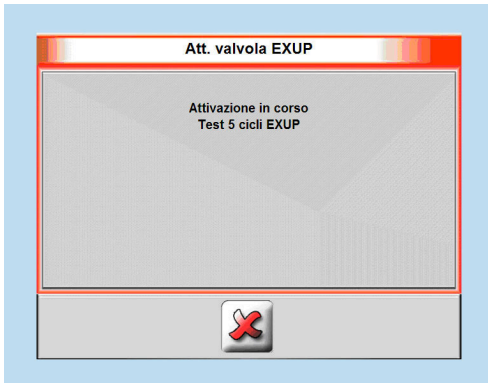
Loosen adjuster ring nut (6) on cable until pulley gets against the limit stop, further loosen it by 1.5 turns.



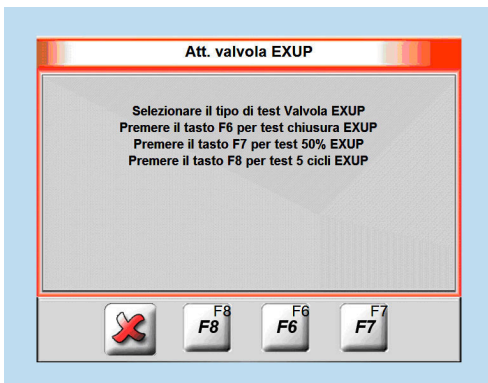
On the DDS, press "F9": to go back to test selection page.



Press "F8" to open function "EXUP 5-cycle test".
The 5-cycle test is performed.



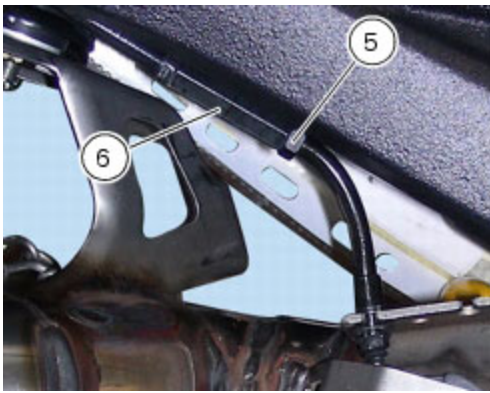
On the DDS, press "F9": to go back to test selection page.



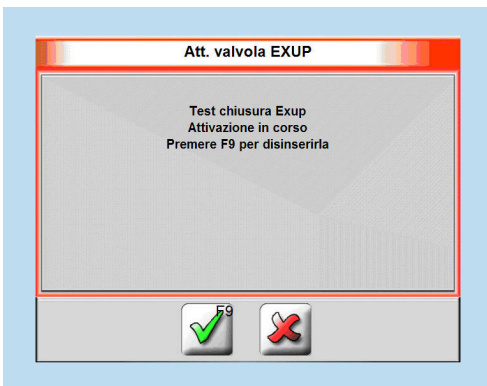
Press "F6" to close the valve that stays closed.



Tighten the adjuster ring nut (6) on cable until you obtain a free play in the range of 0.5 mm to 1 mm measured between limit stop and pulley stop.
Once adjustment is completed, distance between the stop plate welded to the exhaust pipe and the valve pulley front tab must be 0.5 to 1 mm.
Tighten the lock nut (5) to the specified torque (Sect. C 3, [Frame torque settings](#)).



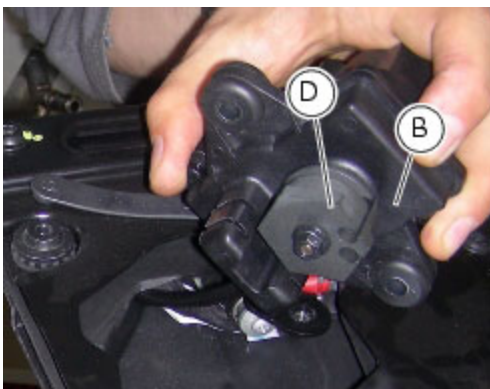
Press button “X” from this DDS page: the activation procedure page closes and system quits without going through test selection.



Disconnect the DDS.

Turn the key to on, let make one complete turn and make sure that the vertex of the hexagon (D) in-between the cable mounting holes is secure/fixed next to the notch (B).

If the test result was positive in the previous step, proceed with the remaining operations. While if result was negative, find failure cause before continuing.



Final checks

Make sure that, upon the following immediate “KEY ON” cycles, the exhaust valve stays in “fully open position”, i.e. the valve pulley rear tab must be resting against the stop plate welded to the exhaust pipe.

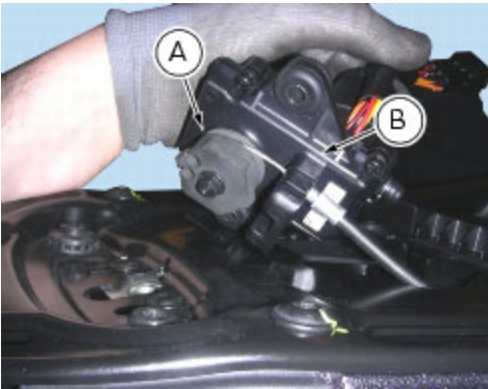


Fitting a new exhaust valve motor with used cable

Insert cable end coming out of the straight sheath length on motor pulley into hole (A) shown.

Insert sheath end into the relevant seat (B) on the motor.

Fit the clip to sheath terminal and make sure that it prevents cable from slipping out.

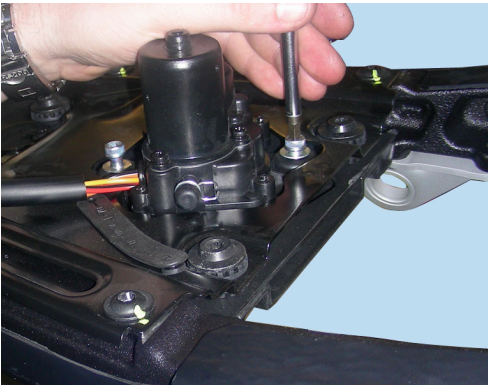


Turn the clip indicated by the arrow to the position shown in the figure.

Route cable in the tunnel provided between plate and number plate holder.

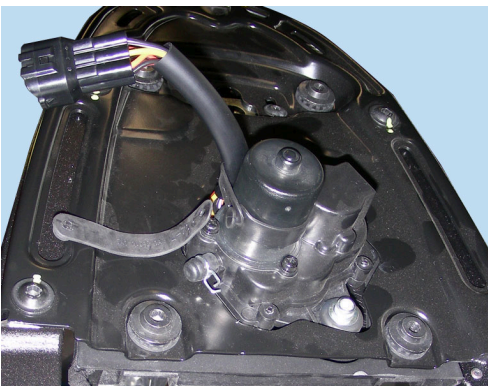
Position and fasten the valve motor to the rear subframe plate by means of two screws and two washers.

Tighten the screws to the specified torque (Sect. C 3, [Frame torque settings](#)).



Fasten the electric cable to the motor central body using a rubber tie L82.

Ensure the cable is not touching the metal plate edge (area indicated by the arrow).

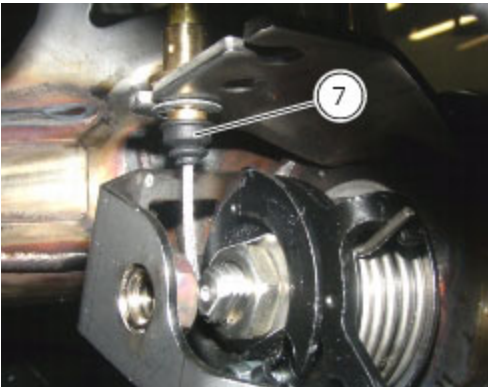
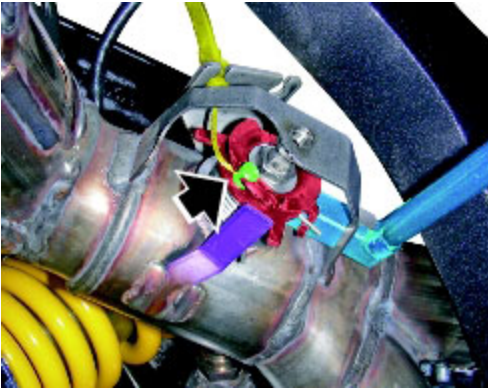


Turn exhaust valve pulley clockwise fully and hold it in this position.

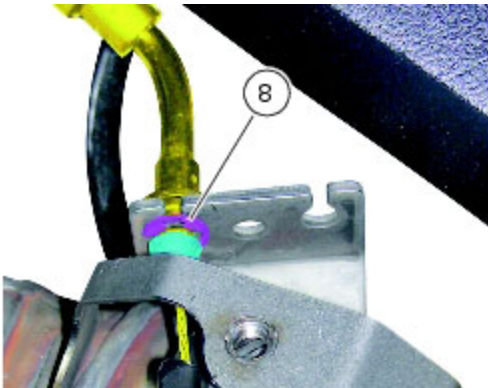
Slide off the dust gaiter (7) from the sheath end.

Fit sheath end into the left-hand hole available on the plate above the valve pulley and refit the dust gaiter.

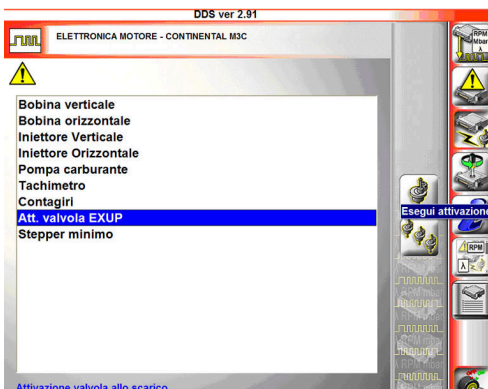
Fit cable end into the pulley hole indicated in the figure and release pulley so as cable is taut again.



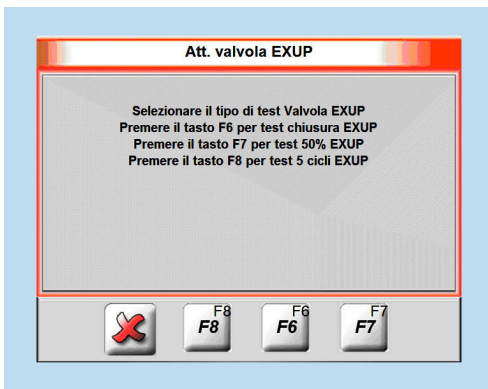
Lock cable end to plate by means of the circlip (8).



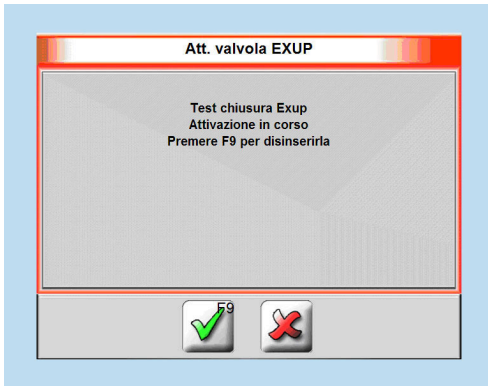
Connect the DDS (Sect. D 5, [Tester power supply](#)) and open the procedure "EXUP valve act."



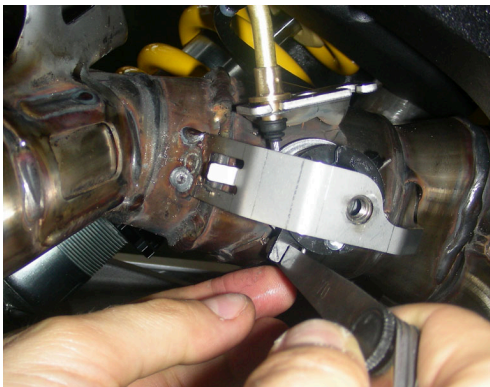
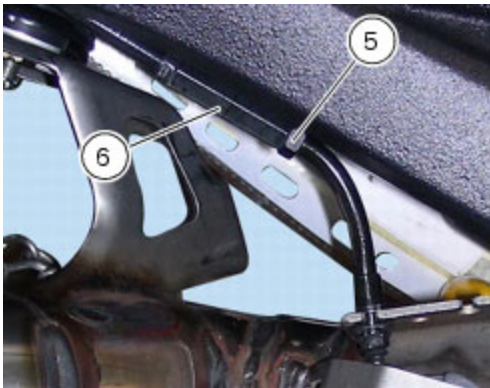
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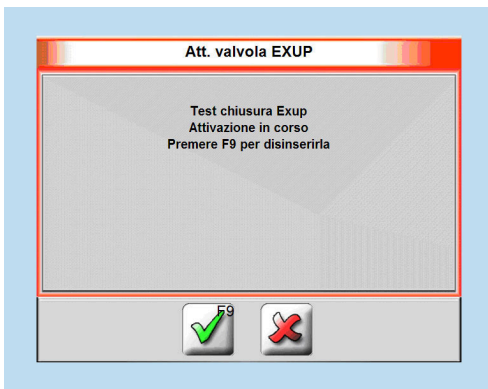
Press "F6" to close the valve that stays closed.



Tighten the adjuster ring nut (6) on cable until you obtain a free play in the range of 0.5 mm to 1 mm measured between limit stop and pulley stop.
Once adjustment is completed, distance between the stop plate welded to the exhaust pipe and the valve pulley front tab must be 0.5 to 1 mm.
Tighten the lock nut (5) to the specified torque (Sect. C 3, [Frame torque settings](#)).



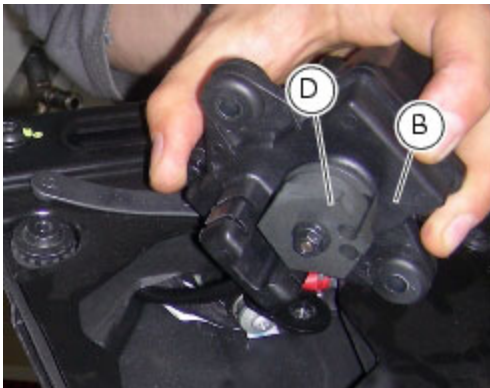
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Final checks

Make sure that, upon the following immediate "KEY ON" cycles, the exhaust valve stays in "fully open position", i.e. the valve pulley rear tab must be resting against the stop plate welded to the exhaust pipe.

